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How to Use This Sheet (Tonight: 3 Rounds, 2 Hours)

Ye sheet tumhara **kal ke exam ka master key** hai. Ek hi PDF, saare 7 chapters, zero filler. Raat ko sirf teen rounds khelo - bas. Round 1 mein formulas aur tricks ratt lo (inhe padhne se hi speed aati hai). Round 2 mein har Important Question ko **pehle khud try karo**, phir solution se mila lo - yahi asli revision hai. Round 3 mein MCQs rapid-fire maaro aur jo galat ho, uske aage star laga do. Subah exam se pehle sirf aakhri page (Exam Morning Power Page) dobara dekho. Agar ye plan follow kiya, toh 80/80 koi badi baat nahi hai.

Round	Kya karna hai	Goal
Round 1 (~40 min)	Saare Formula Tables + Concept Checklists + Speed Tricks	Pura book ka hissa aankh band karke yaad ho jaye - formulas, kab-use-karna, trap warnings.
Round 2 (~60 min)	75 Most Important Questions (chapter-wise)	Book-style best questions with chhote solutions. Khud solve karo, phir check karo.
Round 3 (~40 min)	97 Rapid-Fire MCQs (12-15 per chapter)	4 options, 1 answer, 1-line logic. Galat wale pe star - wo subah dobara dekho.
Subah (~15 min)	Exam Morning Power Page (last page)	Whole book ke 12 killer formulas + exam-hall rules. Fresh dimaag = full marks.

> **Golden rule**
> Pehle KHUD try karo, phir solution dekho - ye 4 ghante tumhe exam hall ka sabse confident banda bana denge. Star system: jo bhi galat ho, aage **star** lagao aur subah sirf starred cheezein dohrao.

Chapter 1: Coordinate Geometry

> **Ek line mein chapter**
> Graph par points ki language: kahan hai (x, y), do points ke beech kitna faasla, midpoint kahan, reflection kaise hota hai. Formulas kam, soch smart - isi chapter mein 3-4-5 triple tumhara best friend ban jata hai.

1.1 Saare Formulas - Easy Bhasha Mein

Har formula ke saath uska **easy matlab** aur ek **tiny example** hai - formula akela kabhi mat rattro, example ke saath yaad karo. Exam mein pehle 2 second sochho kaunsa formula fire karna hai, phir likho.

Formula (ratt lo)	Easy bhasha mein matlab	Tiny example
$d = \sqrt{((x_2-x_1)^2 + (y_2-y_1)^2)}$	Do points ke beech ka seedha faasla. Dono coordinates ka difference lo, square karo, jodo, root nikalo. Bas.	(1, 2) se (4, 6): $\sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$
Origin se $d = \sqrt{(x^2 + y^2)}$	Point (x, y) ka ghar (origin) se faasla. Triples ratt lo: (3,4,5), (5,12,13), (8,15,17) - phir root ki zaroorat hi nahi.	(6, 8): $\sqrt{(36 + 64)} = \sqrt{100} = 10$
Same y? $d = \lvert x_2 - x_1 \rvert$ Same x? $d = \lvert y_2 - y_1 \rvert$	Ek coordinate same hai? Formula chhodo - bas subtract karo. Free marks!	(2, 5) se (9, 5): $\lvert 9 - 2 \rvert = 7$
$M = ((x_1+x_2)/2, (y_1+y_2)/2)$	Midpoint = dono coordinates ka average. "Bisects" dikhe toh yehi.	(1, 2), (5, 8) ka midpoint = (3, 5)
Reverse: $B = 2M - A$ (har coordinate pe)	Ek end + midpoint diya ho toh dusra end = 2×midpoint − end. Midpoint se double karo, ek end minus.	A(1, 3), M(4, 7) → B = (8−1, 14−3) = (7, 11)
x-axis mein reflect: (x, −y) y-axis mein reflect: (−x, y) Origin mein: (−x, −y)	Mirror image: jis axis mein reflect karna hai, USI coordinate ka sign badlo, dusra waise hi rehne do.	(4, −3) in x-axis → (4, 3)
Right angle test: (sabse badi side)^2 = baaki dono ka sum	Pythagoras ka ulta: agar barabar hai toh right angle confirm. Triangle type puchhe toh yehi karo.	3, 4, 5: $5^2 = 25 = 9 + 16$ ✓
Circle: point (x, y) par hai agar $x^2 + y^2 = r^2$	Centre origin wale circle ke liye: x^2+y^2 chhota = andar, barabar = par, bada = bahar.	r = 5, point (3, 4): 9 + 16 = 25 → par hai ✓
On the x-axis → y = 0; on the y-axis → x = 0	Axis par pade point mein dusri coordinate zero hoti hai. Equidistant point maange toh yehi maan ke solve karo.	y-axis par (0, k) lo, dono points se faasla barabar rakho
Right triangle ka area (vertices se) = ½ × leg × leg	Right angle mil jaye toh Heron/box mat karo - dono legs multiply karke aadha. Done.	Legs 8 aur 6 → area = 24 sq units

1.2 Concept Checklist - Ye Sab Aana Hai

- **Quadrant signs:** I (+,+), II (−,+) - upper left, III (−,−), IV (+,−). Axis par point ho toh quadrant NAHI hota.
- **Ordered pair ka order matter karta hai:** (3, 5) ≠ (5, 3). Pehla number x (horizontal), dusra y (vertical).
- **Perpendicular distance:** point se x-axis ka faasla = |y|, y-axis ka = |x|. Sign nahi, magnitude.
- **Collinear points:** teeno ke beech ka bada distance = chhote dono ka sum. Distance formula 3 baar lagao.
- **Parallelogram check:** opposite sides equal (distance formula se) - ya diagonals ka midpoint same.
- **Square/rectangle verify:** sides + diagonals sab check karo; square mein diagonals bhi equal hoti hain.
- **Rectangle ka missing vertex:** opposite corners ka midpoint same hota hai - isse nikalo.
- **Grid-walk vs direct:** galiyon mein chalna (horizontal+vertical) seedhe faasle se LAMBA hota hai, kabhi barabar nahi (except straight line).
- **x^2+y^2 compare karo r^2 se** - andar/par/bahar ka ek-line test.

1.3 Speed Tricks & Trap Alarms

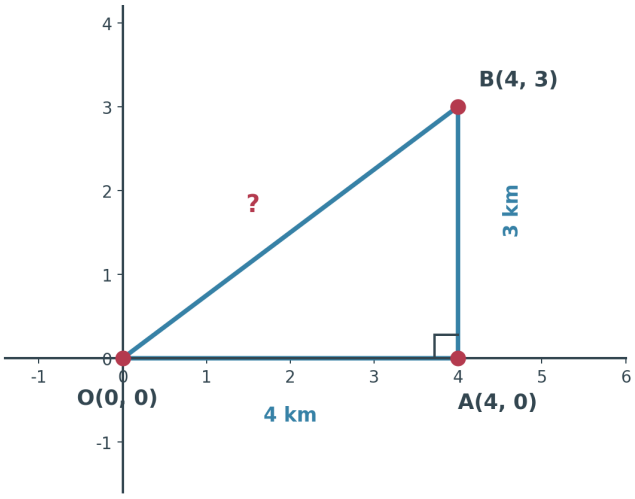
Instant trick	Fire it when...
Same y? Distance = $\|x_2 - x_1\|$. Same x? Distance = $\|y_2 - y_1\|$	Points share a coordinate - skip the formula, just subtract.
Distance from origin = $\sqrt{(x^2 + y^2)}$. Memorise triples: (3,4,5), (6,8,10), (5,12,13), (8,15,17)	"How far from the origin?" - spot the triple, answer in 2 seconds.
Reflect in x-axis: (x, -y). In y-axis: (-x, y). In both: (-x, -y)	Any "mirror image / reflection" wording.
Midpoint = averages of the coordinates; reverse: B = 2M - A	"Midpoint", "bisects", or one endpoint + midpoint given.
Right angle test: (biggest side)^2 = sum of squares of other two	"Prove/verify right angle", triangle classification.
Point on circle centre O radius r $\iff x^2 + y^2 = r^2$	"On/inside/outside the circle" - compare x^2+y^2 with r^2.
"On the x-axis" → y = 0; "on the y-axis" → x = 0	Building coordinates of axis points.

1.4 Most Important Questions (10)

Ye wo questions hain jo **book ke exercises + exam pattern** se seedhe uthaye gaye hain - inka style, numbers aur twist exam mein aane ke sabse zyada chances hain. Solution chhota hai par poora logic covered hai.

IMP Q1 | Figure reading + distance formula (T2/T4)

Q1. A delivery boy starts at home O(0, 0), rides 4 units east to a tea stall at A(4, 0), then rides 3 units north to deliver food at B(4, 3), and finally rides straight back to O. Using the figure: (i) what is the total distance he rode? (ii) when he is at B, how far is he from home (straight line)? (iii) if a drone could fly directly from O to B, how much shorter would the O-to-B trip be compared to his road trip O-A-B?



Step 1. Read the figure: O to A is horizontal with the same y-coordinate, so OA = |4 - 0| = 4 units. A to B is vertical with the same x-coordinate, so AB = |3 - 0| = 3 units. Same-coordinate pairs give free distances.

Step 2. The ride home B to O is the slanted stretch BO. Route makes a right angle at A, so triangle OAB has legs 4 and 3.

Step 3. Distance formula (or the 3-4-5 triple): BO = $\sqrt{((4-0)^2 + (3-0)^2)}$ = $\sqrt{(16 + 9)}$ = $\sqrt{25}$ = 5 units.

Step 4. (i) Total distance ridden = OA + AB + BO = 4 + 3 + 5 = 12 units.

Step 5. (ii) At B, straight-line distance from home = BO = 5 units.

Step 6. (iii) Road O-A-B = 4 + 3 = 7 units; direct O-to-B = 5 units. Saving = 7 - 5 = 2 units.

Answer: (i) 12 units (ii) 5 units (iii) 2 units shorter

Hinglish: *Delivery wala scene ek right triangle hai: east 4, north 3, toh ghar seedha 5 - 3-4-5 triple yaad hai na? Total ride 12 units, par seedhi line sirf 5. Graph paper par ghoomna mehnga hai, drone hi bhej do!*

IMP Q2 | Distance from origin (T4)

Q2. Find the distance of the point (6, 8) from the origin. Which Pythagorean triple is hiding inside this question, and how can the triple help you answer faster next time?

Step 1. Given: point (6, 8). Needed: distance from O(0, 0).

Step 2. Distance from origin = $\sqrt{(x^2 + y^2)}$ = $\sqrt{(6^2 + 8^2)}$.

Step 3. Compute the squares: 36 + 64 = 100.

Step 4. Take the root: $\sqrt{100}$ = 10.

Step 5. The hidden triple is 3-4-5 scaled by 2: 6-8-10. If you spot (6, 8), you can jump straight to 10 without squaring.

Step 6. So the distance is 10 units.

Answer: 10 units; the triple 6-8-10 (which is 2 x (3-4-5)).

Hinglish: *Origin se distance ka short formula: $\sqrt{(x^2 + y^2)}$. 6 aur 8 ko dekhte hi 3-4-5 wala triple yaad karo - dono numbers 2 se multiply ho rahe hain, isliye answer 5 ka 2 guna = 10. Triples ratt lo: 3-4-5, 5-12-13, 8-15-17, 7-24-25. Par dhyan rahe - triple tabhi use karo jab numbers EXACT match karein, warna seedha formula lagao.*

IMP Q3 | Distance formula (T4)

Q3. Find the distance between A(1, 2) and B(4, 6) using (i) the helper-triangle method and (ii) the direct formula. Confirm both give the same value.

Step 1. Given: A(1, 2), B(4, 6). Needed: AB by two methods.

Step 2. Helper triangle: the horizontal leg is $|4 - 1| = 3$ and the vertical leg is $|6 - 2| = 4$.

Step 3. By Pythagoras, $AB = \sqrt{(3^2 + 4^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$.

Step 4. Direct formula: $AB = \sqrt{((4 - 1)^2 + (6 - 2)^2)} = \sqrt{(3^2 + 4^2)} = 5$.

Step 5. Both methods agree: AB = 5 units. The helper triangle simply explains where the formula comes from.

Answer: AB = 5 units by both methods.

Hinglish: *Yahi chapter ka hero question hai. Helper triangle mein humne sirf differences liye - 3 aur 4 - aur wo 3-4-5 triple nikla, isliye answer 5. Formula bhi wahi karta hai, bas ek line mein. Dono ka jawab same aana hi chahiye kyunki formula Pythagoras ka hi shortcut hai. Aage se seedha formula use karo, par agar kabhi doubt ho, helper triangle bana lo.*

IMP Q4 | Reflection chain (T7)

Q4. The point P(4, -3) is reflected in the x-axis to get P_1, and then P_1 is reflected in the y-axis to get P_2. Write the coordinates of P_1 and P_2, and find the length PP_2.

Step 1. Given: P(4, -3) and a chain of two reflections. Needed: images and PP_2.

Step 2. Reflection in the x-axis keeps x and flips the sign of y: $P_1 = (4, 3)$.

Step 3. Reflection in the y-axis keeps y and flips the sign of x: $P_2 = (-4, 3)$.

Step 4. PP_2 joins (4, -3) to (-4, 3): x-difference = -4 - 4 = -8, y-difference = 3 - (-3) = 6.

Step 5. $PP_2 = \sqrt{((-8)^2 + 6^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$.

Step 6. Check with preserved distances: $PP_1 = |3 - (-3)| = 6$ (vertical segment), $P_1P_2 = |(-4) - 4| = 8$ (horizontal segment), and indeed $6^2 + 8^2 = 10^2$, so the numbers are internally consistent.

Answer: $P_1 = (4, 3)$, $P_2 = (-4, 3)$, $PP_2 = 10$ units.

Hinglish: *Reflection ke do simple rules: x-axis mein ungli x-axis par - sirf y ka sign palto. y-axis mein - sirf x ka sign palto. Chain mein dono lagane se (4, -3) pehle (4, 3) bana, phir (-4, 3). Ab distance: 8 aur 6 ka 8-6-10 triple! Aur verification sundar hai: $PP_1 = 6$ vertical, $P_1P_2 = 8$ horizontal, inka Pythagoras bhi 10 hi deta hai. Double confirmation mil gaya.*

IMP Q5 | Right-angle by converse Pythagoras (T6)

Q5. A(1, 1), B(4, 1) and C(4, 5) are three points. Use the distance formula and the converse of the Baudhayana-Pythagoras Theorem to show that angle B is a right angle, and find the area of triangle ABC.

Step 1. Given: three vertices. Needed: prove right angle at B, then area.

Step 2. AB: same y = 1, so AB = |4 - 1| = 3.

Step 3. BC: same x = 4, so BC = |5 - 1| = 4.

Step 4. AC = $\sqrt{((4 - 1)^2 + (5 - 1)^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$.

Step 5. Check the converse: $AC^2 = 25$ and $AB^2 + BC^2 = 9 + 16 = 25$. They are equal, so the angle opposite AC - which is angle B - is a right angle.

Step 6. Area of a right triangle = (1/2) x leg x leg = (1/2) x 3 x 4 = 6 square units.

Step 7. The hidden triple is again 3-4-5.

Answer: Angle B = 90 degrees (since $3^2 + 4^2 = 5^2 = AC^2$); area = 6 square units.

Hinglish: *Converse Pythagoras ka use yaad karo: agar $c^2 = a^2 + b^2$, toh c ke saamne wala angle 90 hai. Yahan AC sabse lambi hai (5), aur $3^2 + 4^2 = 5^2$ nikla, toh B par right angle. Area ke liye right triangle ka bonus: do legs ko base-height maan lo, (1/2)(3)(4) = 6. Coordinate geometry mein area aise hi nikalte hain jab triangle axis-aligned legs rakhta hai.*

IMP Q6 | Reverse midpoint (T8)

Q6. M(-1, 3) is the midpoint of the segment joining A(2, -4) and B(x, y). Find B, and verify your answer by checking that MA = MB.

Step 1. Given: midpoint and one endpoint. Needed: the other endpoint.

Step 2. Midpoint equations: (-1) = (2 + x)/2 and 3 = (-4 + y)/2.

Step 3. Solve the first: multiply by 2: -2 = 2 + x, so x = -4.

Step 4. Solve the second: 6 = -4 + y, so y = 10.

Step 5. So B = (-4, 10).

Step 6. Verify MA: differences (2 - (-1)) = 3 and (-4 - 3) = -7, MA = $\sqrt{(9 + 49)} = \sqrt{58}$.

Step 7. Verify MB: differences (-4 - (-1)) = -3 and (10 - 3) = 7, MB = $\sqrt{(9 + 49)} = \sqrt{58}$. Equal, so B is correct.

Answer: B = (-4, 10), verified since MA = MB = $\sqrt{58}$.

Hinglish: *Midpoint wala equation ulta chalana hai: average diya hai, original nikalna hai. 2 se multiply karo, phir known endpoint minus karo. x ke liye: (-1)(2) - 2 = -4. y ke liye: (3)(2) - (-4) = 10. Verification zaroor karo - MA aur MB dono $\sqrt{58}$ aaye, matlab M sach mein beech mein baitha hai. Bina verification ke answer likhna adhoora hai.*

IMP Q7 | Equidistant point on an axis (T5/T8)

Q7. Find the point on the y-axis that is equidistant from P(-2, 0) and Q(6, 4). Verify your answer.

Step 1. Given: unknown point R on the y-axis. Needed: R equidistant from P and Q.

Step 2. On the y-axis the point has the form R(0, y) - fix the zero first.

Step 3. $RP^2 = (0 - (-2))^2 + (y - 0)^2 = 4 + y^2$.

Step 4. $RQ^2 = (0 - 6)^2 + (y - 4)^2 = 36 + (y - 4)^2$.

Step 5. Set them equal: $4 + y^2 = 36 + y^2 - 8y + 16$.

Step 6. Cancel y^2 from both sides (this is why the equation stays linear and easy): $4 = 52 - 8y$.

Step 7. So $8y = 48$, giving $y = 6$. Hence R = (0, 6).

Step 8. Verify: $RP = \sqrt{(4 + 36)} = \sqrt{40} = 2\sqrt{10}$. $RQ = \sqrt{(36 + (6 - 4)^2)} = \sqrt{(36 + 4)} = \sqrt{40} = 2\sqrt{10}$. Equal, so R is correct.

Answer: R = (0, 6), verified since RP = RQ = $2\sqrt{10}$.

Hinglish: *Y-axis par point = (0, y) - pehle yeh lock karo. Phir dono distances ke SQUARES barabar rakho, root ka jhanjhat khatam. Sabse sundar cheez: y^2 dono taraf cancel ho gaya aur seedha linear equation mil gaya, y = 6. Verify karo: dono distances $2\sqrt{10}$. Aisa equidistant point technically perpendicular bisector par baithta hai - yahan humne uska sirf y-axis wala intersection nikala. $2\sqrt{10}$ ko 6.32 likhne ki zaroorat nahi.*

IMP Q8 | Missing rectangle vertex (T6/T8)

Q8. Three vertices of a rectangle are A(1, 2), B(5, 2) and C(5, 7). Find the fourth vertex D, and verify the rectangle property that its diagonals are equal.

Step 1. Given: three consecutive vertices. Needed: D and the diagonal check.

Step 2. AB is horizontal (y = 2 for both) with length 4; BC is vertical (x = 5 for both) with length 5.

Step 3. In a rectangle the fourth side DA must be vertical (above A) and CD horizontal - so D must sit directly above A at the same height as C.

Step 4. Directly above A(1, 2) means x = 1; at the same height as C means y = 7. So D = (1, 7).

Step 5. Verify with side lengths: DA from (1, 7) to (1, 2) has length 5, CD from (5, 7) to (1, 7) has length 4 - matching the opposite sides.

Step 6. Diagonal $AC = \sqrt{((5 - 1)^2 + (7 - 2)^2)} = \sqrt{(16 + 25)} = \sqrt{41}$.

Step 7. Diagonal $BD = \sqrt{((1 - 5)^2 + (7 - 2)^2)} = \sqrt{(16 + 25)} = \sqrt{41}$. Equal, so the rectangle property holds.

Answer: D = (1, 7); diagonals AC = BD = $\sqrt{41}$, confirming the rectangle.

Hinglish: *Rectangle ka chautha vertex pattern se milta hai: A ke seedha upar, C ke level par. x = 1 (A jaisa), y = 7 (C jaisa). Phir property check - diagonals barabar? Dono $\sqrt{41}$. Yahan $\sqrt{41}$ ko simplify nahi kar sakte (41 prime hai), toh root form hi final hai. Notice karo: har rectangle mein diagonals barabar hote hain - ye test humne parallelogram question (Q23) mein bhi use kiya tha, wahan fail hua tha, yahan pass.*

IMP Q9 | Area of right triangle from coordinates (T12)

Q9. A triangle has vertices O(0, 0), A(8, 0) and B(0, 6). Show that it is right-angled, find its hypotenuse, and compute its area and perimeter.

Step 1. Given: three vertices, two of them on the axes. Needed: right angle, hypotenuse, area, perimeter.

Step 2. OA lies along the x-axis: length 8. OB lies along the y-axis: length 6.

Step 3. The x-axis and y-axis are perpendicular, and OA, OB lie along them, so the angle at O is 90 degrees.

Step 4. Hypotenuse AB = $\sqrt{(8^2 + 6^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$ (the 6-8-10 triple).

Step 5. Area = (1/2) x OA x OB = (1/2) x 8 x 6 = 24 square units.

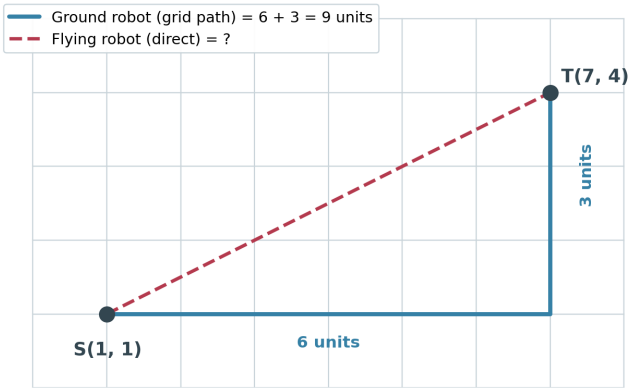
Step 6. Perimeter = 8 + 6 + 10 = 24 units. A charming coincidence: area 24 and perimeter 24.

Answer: Right angle at O; hypotenuse AB = 10; area = 24 sq units; perimeter = 24 units.

Hinglish: *Jab triangle ke do sides axes par baithe hon, right angle free mein milta hai - axis parallel hi perpendicular hote hain. Phir 8-6-10 triple se hypotenuse 10. Area = 24, perimeter bhi 24 - dono number same! Aisi coincidences questions mein jaan-bujh ke rakhi jaati hain, toh apne practice questions mein bhi aisi numbers chuno toh checking easy hoti hai.*

IMP Q10 | Grid-walk vs direct distance (T11/T4)

Q10. A delivery robot moves on a grid of unit squares. It must travel from station S(1, 1) to station T(7, 4). (i) What is the length of the shortest path using only horizontal and vertical moves? (ii) What is the straight-line distance? (iii) A second robot flies directly. How much distance does it save compared to the ground robot?



Step 1. Given: grid unit 1, S(1, 1) to T(7, 4). Needed: street length, direct length, saving.

Step 2. Horizontal gap = 7 - 1 = 6 units; vertical gap = 4 - 1 = 3 units.

Step 3. Ground robot (only axis moves): 6 + 3 = 9 units. Every such path has exactly this length - no order of moves can beat it.

Step 4. Flying robot (straight line): $\sqrt{(6^2 + 3^2)} = \sqrt{(36 + 9)} = \sqrt{45} = 3\sqrt{5}$ (approximately 6.71 units).

Step 5. Saving = $9 - 3\sqrt{5}$ (exactly), approximately 9 - 6.71 = 2.29 units.

Step 6. Keep the exact answer $9 - 3\sqrt{5}$ in the final statement; the decimal is only for intuition.

Answer: (i) 9 units (ii) 3√5 units (approx 6.71) (iii) saving 9 - 3√5 approx 2.29 units.

Hinglish: *Grid-walk ka funda: horizontal steps + vertical steps hamesha fix hote hain (yahan 6 + 3 = 9), chahe kisi bhi order mein chalo. Direct distance Pythagoras se 3√5. Saving 9 - 3√5 approx 2.3 units. Ye question Q26 (city auto) ka bhai hai - wahan 200 m blocks the, yahan 1 unit. Same skeleton, naye numbers - yahi 'question structure' samajhna original practice hai, ratna nahi.*

1.5 Rapid-Fire MCQs (14)

Rapid fire: option padho, dimaag mein answer banao, phir Answer line se check karo. Galat hua toh Answer line ka logic padho - wahi 1 line exam mein marks bachayegi.

1. The point (-7, 4) lies in which quadrant?

(A) Quadrant I **(B)** Quadrant II **(C)** Quadrant III **(D)** Quadrant IV

Ans: (B) · x = -7 is negative and y = +4 is positive. The sign pattern (-, +) is exactly the definition of Quadrant II (above the x-axis, left of the y-axis).

2. The perpendicular distance of the point (-4, 9) from the y-axis is:

(A) 9 units **(B)** 4 units **(C)** -4 units **(D)** √97 units

Ans: (B) · Distance from the y-axis is given by the size of the x-coordinate: |x| = |-4| = 4. Distances are never negative.

3. The distance between (1, -2) and (7, 6) is:

(A) 10 units **(B)** 12 units **(C)** √84 units **(D)** 14 units

Ans: (A) · Differences: 7 - 1 = 6 and 6 - (-2) = 8. Distance = √(36 + 64) = √100 = 10, the 6-8-10 triple.

4. If P(a, b) lies in Quadrant II, then the point (-a, -b) lies in:

(A) Quadrant I **(B)** Quadrant II **(C)** Quadrant III **(D)** Quadrant IV

Ans: (D) · In Q II, a is negative and b is positive. So -a is positive and -b is negative: signs (+, -), which is Quadrant IV. Flipping both signs reflects the point through the origin (a half-turn).

5. The reflection of the point (-3, -5) in the y-axis is:

(A) (3, -5) **(B)** (3, 5) **(C)** (-3, 5) **(D)** (-3, -5)

Ans: (A) · Reflection in the y-axis flips only the sign of x: (-3, -5) becomes (3, -5). The y-coordinate is untouched.

6. The midpoint of the segment joining (-3, 8) and (5, -2) is:

(A) (1, 3) **(B)** (2, 6) **(C)** (1, 5) **(D)** (4, 3)

Ans: (A) · Midpoint = ((-3 + 5)/2, (8 + (-2))/2) = (2/2, 6/2) = (1, 3).

7. The distance of (5, -12) from the origin is:

(A) 7 units **(B)** 13 units **(C)** 17 units **(D)** 169 units

Ans: (B) · $\sqrt{(25 + 144)} = \sqrt{169} = 13$. The 5-12-13 triple, again with a harmless minus sign.

8. The perimeter of the triangle with vertices (0, 0), (4, 0), (0, 3) is:

(A) 7 units **(B)** 12 units **(C)** 10 units **(D)** 6 units

Ans: (B) · Legs are 4 and 3; hypotenuse = 5 (3-4-5). Perimeter = 4 + 3 + 5 = 12.

9. The area of the triangle with vertices (0, 0), (8, 0) and (0, 6) is:

(A) 48 sq units **(B)** 24 sq units **(C)** 14 sq units **(D)** 10 sq units

Ans: (B) · Right angle at the origin with legs 8 and 6. Area = (1/2)(8)(6) = 24.

10. Assertion (A): The distance of a point (x, y) from the origin is $\sqrt{(x^2 + y^2)}$. Reason (R): The Baudhayana-Pythagoras Theorem gives hypotenuse = $\sqrt{}$(sum of squares of legs). Choose the correct option:

(A) Both A and R are true and R explains A **(B)** Both A and R are true but R does not explain A **(C)** A is true, R is false **(D)** A is false, R is true

Ans: (A) · The distance formula from the origin is exactly Pythagoras applied to the legs x and y. The assertion is true and the reason is the actual justification behind it.

11. The point (x, 0) is equidistant from (2, 3) and (6, 3). Then x equals:

(A) 4 **(B)** 2 **(C)** 8 **(D)** 3

Ans: (A) · $(x - 2)^2 + 9 = (x - 6)^2 + 9$. The 9s cancel: $(x - 2)^2 = (x - 6)^2$, so x - 2 = plus/minus (x - 6). The negative-sign branch gives a contradiction (0 = -8... no solution from that branch), the positive-sign branch gives x - 2 = x - 6? Also impossible - so take x - 2 = -(x - 6): 2x = 8, x = 4. Geometrically, x = 4 is the midpoint of 2 and 6, directly below the midpoint of the two given points.

12. Statement: If the two points in a distance calculation are interchanged, the computed distance changes. Is the statement TRUE or FALSE, and why?

(A) TRUE, because x and y swap places **(B)** FALSE, because the differences are squared, which erases the sign change **(C)** TRUE, because the formula is order-dependent **(D)** FALSE, because distance is always zero

Ans: (B) · Swapping the points changes (x2 - x1) into (x1 - x2) = negative of it. But squaring makes both equal. Hence the distance is unchanged.

13. A point whose x-coordinate is 0 must lie:

(A) on the x-axis **(B)** on the y-axis **(C)** in Quadrant II **(D)** at the origin

Ans: (B) · x = 0 means zero sideways distance from the y-axis, so the point sits on the y-axis - at (0, y) for some y. It is the origin only if additionally y = 0.

14. For the right triangle with vertices (0, 0), (6, 0) and (0, 8), the point equidistant from all three vertices is:

(A) (3, 4) **(B)** (3, 5) **(C)** (4, 3) **(D)** (2, 3)

Ans: (A) · Hypotenuse joins (6, 0) to (0, 8); its midpoint is (3, 4). In a right triangle the midpoint of the hypotenuse is equidistant from all three vertices (distance 5 here: 3-4-5 again).

Chapter 2: Linear Equations in One and Two Variables

> **Ek line mein chapter**
> Duniya ke har "fixed + per-unit" scene ko ek line y = ax + b mein likhna. Bill ho, age ho, tanki ho ya matchstick pattern - sab isi ek shape mein aata hai. Slope aur intercept padhna seekh lo, chapter khatam.

2.1 Saare Formulas - Easy Bhasha Mein

Har formula ke saath uska **easy matlab** aur ek **tiny example** hai - formula akela kabhi mat rattro, example ke saath yaad karo. Exam mein pehle 2 second sochho kaunsa formula fire karna hai, phir likho.

Formula (ratt lo)	Easy bhasha mein matlab	Tiny example
y = ax + b	**a** = har unit ka rate (slope), **b** = shuruati/fixed amount. "Rs 300 joining + Rs 60 per session" → y = 60x + 300.	y = 60x + 300; x = 5 sessions → y = 600
Slope from 2 points: a = $(y_2-y_1)/(x_2-x_1)$	Per step kitna badla. Upar utha toh plus, neeche gira toh minus.	(1, 4), (3, 10): a = (10−4)/(3−1) = 3
Do bills diya ho? Equations MINUS karo	Dono equations ko subtract karo - fixed wala (b) cancel ho jata hai, rate (a) khud nikal aata hai. Phir b wapas nikaalo.	y = ax + b ke 2 points → pehle a, phir b
Pattern: T(n) = kn + c, jahan c = (first term) − k	Har step mein utna hi add ho (k)? Toh linear pattern. c nikaalne ke liye pehle term se k ghatao.	4, 9, 14, ...: k = 5, c = 4−5 = −1 → T(n) = 5n − 1
y-intercept = (0, b); x-intercept: y = 0 rakh ke solve	Line jahan y-axis ko kaate = (0, b). x-axis ka point chahiye toh y ko zero maan lo.	y = 2x + 8: y-cut (0, 8), x-cut (−4, 0)
Bada |a| = steeper line; same a = parallel lines	Slope compare karo: jiska |a| bada, wo zyada khadi. a same ho toh kabhi milengi nahi (parallel).	y = 3x − 1 steeper than y = x + 9
Growth: add/earn → +a. Decay: spend/drain → −a	Scene padho: paisa jud raha hai toh slope plus, fat raha hai toh minus. Decay mein bhi formula same.	Bela: 500 − 30x (decay); tanki: 900 − 45t
p(x) ki value: x ki jagah number daal do	Polynomial machine hai - input daalo, output lo. p(x) = 3x − 7 at x = 5: 15 − 7 = 8.	p(−2) for 4x − 9 = −8 − 9 = −17
Degree = highest power. Linear = degree 1	2x + 9 linear, x^2 + 1 quadratic, x^3 cubic. Constant polynomial ka degree 0.	$4y^3$ + y: degree 3, cubic

2.2 Concept Checklist - Ye Sab Aana Hai

- **Variable pehchano:** jo cheez badal rahi hai (sessions, km, months) wahi x hai; jo puchha hai wo y.
- **Fixed fee kabhi x se multiply nahi hoti** - wo b hai, chahe kuch bhi ho.
- **Do conditions = do equations.** "Sum 50 aur difference 8" jaise scenes minus/karne par solve hote hain.
- **Age problems:** "Mother 4 times daughter" → m = 4d; saalon baad dono mein +k add hota hai.
- **Coin/money problems:** count alag, value alag - 5x + 10y type equation banao.
- **Matchstick patterns:** pehla figure alag (c), har naya figure fixed adds (k) → T(n) = kn + c.
- **Line ki graph reading:** (0, b) se shuru, slope a se chadho/utro. Har point (x, y) line par hai agar equation satisfy ho.
- **Two lines same equation = same line** (infinite solutions); same a different b = parallel (no meeting).
- **C ≠ F kabhi barabar nahi hota?** Hota hai! F = 1.8C + 32 mein C = F rakho: C = −40. Yaad rakhna, HOTS favourite.

2.3 Speed Tricks & Trap Alarms

Instant trick	Fire it when...
Fixed fee + rate × quantity → y = ax + b (b = fixed part, a = per-unit rate)	"Rs 300 joining + Rs 60 per session" style wording.
Two observations? Subtract the equations - b cancels, a pops out	Fit y = ax + b from two bills or data points.
Pattern adds k every step → T(n) = kn + c; c = (first term) - k	Matchstick, tile and money patterns.
Slope a = Δy/Δx; bigger |a| = steeper; same a = parallel	"Which line is steeper / are they parallel?"
Changing b shifts the line up/down; changing a pivots it about (0, b)	"What happens to the graph if ...?"
Spends / loses / drains → minus slope (decay). Adds / earns / fills → plus (growth)	Growth vs decay identification.
y-intercept = (0, b); x-intercept: put y = 0, solve	"Where does the line cut the axes?"

2.4 Most Important Questions (10)

Ye wo questions hain jo **book ke exercises + exam pattern** se seedhe uthaye gaye hain - inka style, numbers aur twist exam mein aane ke sabse zyada chances hain. Solution chhota hai par poora logic covered hai.

IMP Q1 | Word problem: sum and difference (T5)

Q1. The sum of two numbers is 50, and one of them is 8 more than the other. Find the numbers.

Step 1. Let the smaller number be x. Then the larger is x + 8.

Step 2. Sum condition: x + (x + 8) = 50.

Step 3. Simplify: 2x + 8 = 50, so 2x = 42, giving x = 21.

Step 4. The numbers are 21 and 29.

Step 5. Check in words: 21 + 29 = 50 and 29 - 21 = 8. Both conditions hold.

Answer: The numbers are 21 and 29.

Hinglish: *Chota number x maan lo, bada apne aap x + 8 ban jaata hai. Sum equation se x = 21. Aur LAST STEP yaad rakho: dono numbers report karo - sirf 'x = 21' likha toh aadha answer hai. Words mein wapas check karo, equation mein hi nahi.*

IMP Q2 | Fee structure modelling (T5/T8)

Q2. A badminton academy charges a joining fee of Rs 300 plus Rs 60 for every session attended. (i) Write the total cost for n sessions. (ii) Find the cost for 12 sessions. (iii) A player paid Rs 780 in total. How many sessions did she attend?

Step 1. Cost = fixed part + variable part = 300 + 60n.

Step 2. For 12 sessions: 300 + 60(12) = 300 + 720 = Rs 1020.

Step 3. Reverse: 300 + 60n = 780.

Step 4. Subtract 300: 60n = 480, so n = 8.

Step 5. Check: 300 + 60(8) = 300 + 480 = 780. Correct - she attended 8 sessions.

Answer: (i) 300 + 60n (ii) Rs 1020 (iii) 8 sessions.

Hinglish: *Fixed + rate x count - yahi har bill ka formula hai. Aage jaane ke liye substitute, peeche aane ke liye solve. 780 mein se 300 hataya toh sessions ka khel bacha: 480/60 = 8.*

IMP Q3 | Linear decay money (T8)

Q3. Bela has Rs 500 and spends Rs 30 every day. (i) Write the amount left after n days. (ii) After how many days will she be left with Rs 200? (iii) What is the amount left after 15 days, and why does the model stay valid only up to 10 days of reaching zero? Actually find the exact day her money runs out.

Step 1. Amount(n) = 500 - 30n.

Step 2. Set 500 - 30n = 200: 30n = 300, n = 10 days.

Step 3. After 15 days: 500 - 30(15) = 500 - 450 = Rs 50.

Step 4. Running out: 500 - 30n = 0 gives n = 500/30 = 16.67 days. Money reaches zero partway through the 17th day.

Step 5. So the linear model describes her cash only until about 16.67 days; beyond that the amount cannot go below 0.

Answer: (i) 500 - 30n (ii) after 10 days (iii) Rs 50 after 15 days; money hits zero during the 17th day (n = 500/30 = 16.67), where the model ends.

Hinglish: *Decay question mein teen sawaal hote hain: kitna bacha (substitute), kab itna bacha (solve), kab khatam (zero set karo). Yahan 16.67 din - matlab 17ve din ke beech mein khatam. Model ki limit batana bhi answer ka hissa hai - paisa negative nahi ho sakta!*

IMP Q4 | Piecewise-style fare (T8)

Q4. A taxi charges Rs 40 for the first 3 km (and any part of it), then Rs 18 for every additional km. (i) Find the fare for a 12 km trip. (ii) Write the fare for a trip of n km where n is at least 3. (iii) For what distance is the fare Rs 148?

Step 1. Beyond 3 km, chargeable extra distance = (n - 3) km.

Step 2. Fare for 12 km: 40 + 18 x (12 - 3) = 40 + 162 = Rs 202.

Step 3. General: F(n) = 40 + 18(n - 3) = 18n - 54 + 40 = 18n - 14, valid for n at least 3.

Step 4. Set 18n - 14 = 148: 18n = 162, n = 9 km.

Step 5. Check with the original description: 40 for the first 3 km plus 18 x 6 = 108 for the next 6 km: total 148. Correct.

Answer: (i) Rs 202 (ii) F(n) = 18n - 14 for n at least 3 (iii) 9 km.

Hinglish: *PDF ke auto-fare jaisa hi question, bas numbers naye. Formula banane ke liye (n - 3) ka idea zaroori hai - pehle 3 km ka paisa already fixed 40 mein hai. Simplify karke 18n - 14 milta hai aur reverse solve se 9 km. Hamesha original description se verify karo.*

IMP Q5 | Age problem (T5)

Q5. A mother is 4 times as old as her daughter. After 6 years, the sum of their ages will be 82 years. Find their present ages.

Step 1. Let the daughter be x now; the mother is 4x.

Step 2. After 6 years: daughter x + 6, mother 4x + 6.

Step 3. Sum: (x + 6) + (4x + 6) = 82.

Step 4. Simplify: 5x + 12 = 82, so 5x = 70, x = 14.

Step 5. Present ages: daughter 14, mother 56.

Step 6. Check: after 6 years they are 20 and 62, sum 82. Correct.

Answer: Daughter 14 years, mother 56 years.

Hinglish: *Age questions mein 'after' ka matlab dono ke age mein addition. 5x + 12 = 82 se x = 14. Check zaroori: 20 + 62 = 82. Mother 4x isliye 56 - equation se nahi, definition se.*

IMP Q6 | Fit y = ax + b (T9)

Q6. A learning platform charges a fixed monthly fee plus a per-module charge. A student observed that accessing 10 modules cost Rs 400 and accessing 14 modules cost Rs 500. Assuming the bill is y = ax + b, find a and b, and interpret both numbers.

Step 1. Substitute both observations into y = ax + b: 400 = 10a + b and 500 = 14a + b.

Step 2. Subtract the first from the second: 100 = 4a, so a = 25.

Step 3. Back-substitute into the first: 400 = 10(25) + b = 250 + b, so b = 150.

Step 4. The relationship is y = 25x + 150.

Step 5. Interpretation: a = 25 is the charge per module (Rs 25 per module); b = 150 is the fixed monthly fee (Rs 150 even with zero modules).

Step 6. Verify with the second point: 25(14) + 150 = 350 + 150 = 500. Correct.

Answer: a = 25 (per-module rate), b = 150 (fixed fee); y = 25x + 150.

Hinglish: *Do observation, do equation, ek SUBTRACTION - b khatam, a seedha mil gaya (25). Phir wapas jao, b nikaalo (150). Interpretation zaroor likho: 25 = per-module charge, 150 = fixed fee. Ye interpretation wali line hi examiner ko dikhati hai ki tumne samjha hai, sirf calculation nahi kiya.*

IMP Q7 | Temperature scales (T9)

Q7. The relation between Celsius C and Fahrenheit F has the linear form F = aC + b. Water melts at 0 degrees Celsius and 32 degrees Fahrenheit, and boils at 100 degrees Celsius and 212 degrees Fahrenheit. Find a and b.

Step 1. First data point: C = 0, F = 32 gives 32 = a(0) + b, so b = 32 directly - a gift.

Step 2. Second data point: C = 100, F = 212 gives 212 = 100a + 32.

Step 3. So 100a = 180, giving a = 1.8 (which is 9/5).

Step 4. The relation is F = 1.8C + 32, the famous conversion formula.

Step 5. Verification: C = 100 gives 180 + 32 = 212. Correct.

Answer: a = 1.8 (= 9/5), b = 32; F = 1.8C + 32.

Hinglish: *Yahan data ki kripa se b PEHLE mil gaya (zero input pe output hi b hai!). Phir sirf ek equation bachi, a = 1.8. Ye formula tum pehle se jaante ho - ab tumne use DERIVE kiya hai. Fitting ka asli maza yahi hai.*

IMP Q8 | Matchstick pattern (T6/T7)

Q8. Squares are made with matchsticks placed in a row, each new square sharing one side with the previous one. The counts are 4, 7, 10, ... (i) How many matchsticks for n squares? (ii) For 22 squares? (iii) Can exactly 100 matchsticks form such a row of squares? Justify.

Step 1. Sequence 4, 7, 10 has constant difference 3: rule M(n) = 3n + c. From stage 1: 3 + c = 4, so c = 1. Rule: M(n) = 3n + 1.

Step 2. For 22 squares: M(22) = 66 + 1 = 67 matchsticks.

Step 3. Reachability: set 3n + 1 = 100, giving 3n = 99, n = 33.

Step 4. n = 33 is a whole number, so YES - a row of 33 squares uses exactly 100 matchsticks.

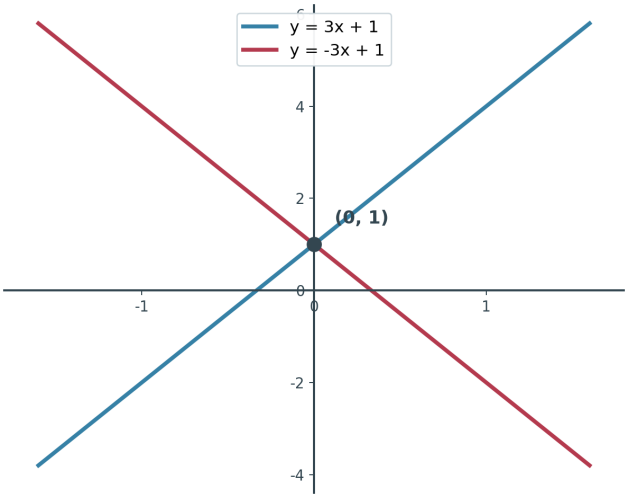
Step 5. The justification is the integrality of n: patterns exist only at whole-number stages.

Answer: (i) 3n + 1 (ii) 67 (iii) Yes - stage n = 33 uses exactly 100 matchsticks.

Hinglish: *3n + 1 wala rule - ek square 4, har naya square sirf 3 extra le raha hai kyunki ek side share hoti hai. 'Can 100 form a stage?' ka matlab: solve karo aur INTEGRITY check karo. Yahan n = 33 whole number - YES. Agar 33.33 aata toh answer 'No, kabhi nahi' hota - aur wahi asli answer hota.*

IMP Q9 | Compare two lines (T11/T12)

Q9. Draw-free reasoning: compare the graphs of y = 3x + 1 and y = -3x + 1. State their common features, their differences, and the geometric relationship between the two lines.



Step 1. Common feature: same b = 1, so both cut the y-axis at the same point (0, 1).

Step 2. Difference: slopes +3 and -3 - equal in size, opposite in sign.

Step 3. Positive slope: as x grows, y grows (uphill). Negative slope: as x grows, y falls (downhill).

Step 4. Geometric relationship: the two lines are mirror images of each other in the y-axis. Reason: replacing x by -x in y = 3x + 1 gives y = -3x + 1 - reflection in the y-axis flips the sign of x.

Step 5. They are NOT parallel (slopes differ), so they cross - exactly at the shared point (0, 1).

Answer: Both pass through (0, 1); slopes +3 and -3 make them mirror images in the y-axis, crossing at (0, 1) - not parallel.

Hinglish: *Same intercept, opposite slopes: dono lines (0, 1) mein milti hain aur wahan se ek upar, ek neeche jaati hain - y-axis ke aaine mein twin. 'Replace x by -x' wala reflection argument coordinate geometry ka classic move hai - Chapter 1 ke reflections se seedha connection.*

IMP Q10 | Wire rectangle area function (T14)

Q10. A wire of length 36 cm is bent into a rectangle. Let the length be x cm. (i) Write the area as a function of x. (ii) Find ALL lengths x that give an area of 77 sq cm. (iii) Which single length gives the largest area, and what is that area?

Step 1. Perimeter 36 means half-perimeter 18: length + width = 18. Width = 18 - x.

Step 2. Area A(x) = x(18 - x) = 18x - x^2 - a quadratic function (the PDF builds exactly this in its Example 3 with a 20 cm wire).

Step 3. Set x(18 - x) = 77: x^2 - 18x + 77 = 0. Look for two numbers with product 77 and sum 18: 7 and 11.

Step 4. So x = 7 or x = 11. Both are genuine: a 7 x 11 rectangle and an 11 x 7 rectangle are the SAME rectangle turned around - one answer in two descriptions.

Step 5. Largest area: test symmetric candidates. x = 9 gives 9 x 9 = 81; x = 8 gives 8 x 10 = 80; x = 10 gives 10 x 8 = 80. The square (x = 9) wins with area 81.

Step 6. Structure noticed: areas rise toward x = 9 and fall after it - symmetric about the square.

Answer: (i) A(x) = 18x - x^2 (ii) x = 7 or 11 (the same rectangle, two descriptions) (iii) x = 9 gives the maximum area 81 sq cm.

Hinglish: *Ye PDF ke wire example ka bada bhai. Area quadratic nikla (18x - x^2) - chapter linear hai par evaluation quadratic ki bhi practice hai (Ex 2.2 mein tha). 77 ke liye do answers (7 aur 11) ek hi rectangle ke do naam hain - symmetry! Aur square (9 x 9 = 81) max deta hai - ye observation Grade 10 mein formally aayega, aaj ke liye table-test kaafi hai.*

2.5 Rapid-Fire MCQs (14)

Rapid fire: option padho, dimaag mein answer banao, phir Answer line se check karo. Galat hua toh Answer line ka logic padho - wahi 1 line exam mein marks bachayegi.

1. The coefficient of x in the expression 5x - 7 is:

(A) 5 **(B)** -7 **(C)** 5x **(D)** 1

Ans: (A) · The coefficient is the number multiplying the variable x, which is 5. The -7 is the constant term.

2. The degree of the polynomial 4 - 3y + y^2 is:

(A) 1 **(B)** 2 **(C)** 3 **(D)** 0

Ans: (B) · The powers present are 0 (the 4), 1 (the -3y) and 2 (the y^2). The highest is 2, so the degree is 2.

3. The degree of the constant polynomial -9 is:

(A) -9 **(B)** 0 **(C)** 1 **(D)** undefined

Ans: (B) · Any constant c can be written as c x x^0, and x^0 = 1. So the degree (power of the variable) is 0.

4. The value of 4x - 9 at x = -2 is:

(A) -17 **(B)** 17 **(C)** -1 **(D)** 1

Ans: (A) · 4(-2) - 9 = -8 - 9 = -17. Both terms are negative here.

5. The machine y = 5x + 2 gives output 27. The input x is:

(A) 5 **(B)** 4 **(C)** 6 **(D)** 25

Ans: (A) · 5x + 2 = 27 gives 5x = 25, so x = 5.

6. Two numbers add to 46, and one of them is 6 more than the other. The numbers are:

(A) 20 and 26 **(B)** 23 and 23 **(C)** 18 and 28 **(D)** 21 and 25

Ans: (A) · Let them be x and x + 6: 2x + 6 = 46 gives x = 20. Numbers 20 and 26.

7. A sequence begins 4, 9, 14, 19, ... Its nth-term rule is:

(A) 5n - 1 **(B)** 5n + 1 **(C)** 4n **(D)** n + 3

Ans: (A) · The constant difference is 5, so the rule is 5n + c. Stage 1: 5 + c = 4 gives c = -1. Rule: 5n - 1. Check stage 2: 10 - 1 = 9. Correct.

8. A service costs Rs 90 for 2 units and Rs 210 for 6 units. In the model y = ax + b:

(A) a = 30, b = 30 **(B)** a = 30, b = 90 **(C)** a = 45, b = 0 **(D)** a = 35, b = 20

Ans: (A) · 210 - 90 = 120 = 4a, so a = 30. Back-substitute: 90 = 2(30) + b = 60 + b, b = 30. Model: y = 30x + 30.

9. In the conversion F = 1.8C + 32, the number 1.8 tells us that:

(A) a 1 degree rise in Celsius gives a 1.8 degree rise in Fahrenheit **(B)** Fahrenheit is always 1.8 more than Celsius **(C)** water boils at 1.8 degrees **(D)** the scales meet at 1.8

Ans: (A) · The slope is the rate of change: each extra degree Celsius lifts the Fahrenheit reading by 1.8. (Option B confuses slope with a constant offset - the offset is 32, not 1.8.)

10. The slope of the line 3y = 6x + 12 is:

(A) 6 **(B)** 2 **(C)** 3 **(D)** 12

Ans: (B) · Divide by 3 first: y = 2x + 4. The slope is 2. Reading the raw coefficient 6 before isolating y is the classic trap.

11. Which of these lines passes through the origin?

(A) y = 5x **(B)** y = 5x + 1 **(C)** y = x - 3 **(D)** y = 2

Ans: (A) · A line passes through (0, 0) exactly when its constant term b is 0. Only y = 5x has b = 0.

12. The lines y = 4x - 3 and y = 4x + 7 are:

(A) perpendicular **(B)** parallel **(C)** the same line **(D)** intersecting at one point

Ans: (B) · Same slope 4, different intercepts (-3 vs 7): parallel lines that never meet.

13. The model h(t) = 3 - 0.5t describes:

(A) linear growth **(B)** linear decay **(C)** no change **(D)** quadratic behaviour

Ans: (B) · The rate of change (slope) is -0.5: each month the height drops by 0.5. Negative slope means decay.

14. The line y = 3x + 6 and the two coordinate axes form a triangle. Its area is:

(A) 6 sq units **(B)** 12 sq units **(C)** 3 sq units **(D)** 18 sq units

Ans: (A) · y-intercept (0, 6); x-intercept: 3x + 6 = 0 gives x = -2, point (-2, 0). Right triangle with legs 6 and 2: area = (1/2)(6)(2) = 6.

Chapter 3: Number Systems and Exponents

Ek line mein chapter

Numbers ki family tree (N, Z, Q, R), recurring decimals ko fraction banana, surds ko rationalise karna aur exponent laws. Proof wale questions (√2, √3, √5 irrational) pakke aate hain - skeleton ratt lo.

3.1 Saare Formulas - Easy Bhasha Mein

Har formula ke saath uska **easy matlab** aur ek **tiny example** hai - formula akela kabhi mat rattro, example ke saath yaad karo. Exam mein pehle 2 second sochho kaunsa formula fire karna hai, phir likho.

Formula (ratt lo)	Easy bhasha mein matlab	Tiny example
$N \subset Z \subset Q \subset R$	Counting numbers → unme negatives jude → unme fractions jude → unme irrationals jude. Har naya set purana cover karta hai.	$-3 \in Z$, $2/7 \in Q$, $\sqrt{2} \in R$ par Q se bahar
Terminating test: fraction ko REDUCE karo, denominator sirf $2^m \times 5^n$ ho toh terminate	Pehle simplify karo (33/125 → already clean), phir denominator ke prime factors dekho. 3, 7, 11 aaye = non-terminating repeating.	$17/312 \rightarrow 312 = 2^3 \times 3 \times 13 \rightarrow$ repeating
Pure recurring: 0.abababab... = (block)/(99...9)	Jitne digits repeat, utne 9s denominator mein. 0.777... = 7/9; 0.454545... = 45/99 = 5/11.	0.666... = 6/9 = 2/3
Mixed recurring: (sab digits – non-repeat)/(9...90...0)	Upar: poori decimal ke digits minus jo pehle the. Neeche: 9s (repeating ke liye) + 0s (non-repeating ke liye).	0.4777... = (47−4)/90 = 43/90
$\sqrt{a} \times \sqrt{b} = \sqrt{(ab)}$; $\sqrt{a}/\sqrt{b} = \sqrt{(a/b)}$	Same root ke surds multiply/divide ho toh andar jodo/kaato. Pehle simplify karna best: $\sqrt{72} = 6\sqrt{2}$.	$\sqrt{2} \times \sqrt{8} = \sqrt{16} = 4$
Rationalise: conjugate se multiply karo	Neeche $\sqrt{a} \pm \sqrt{b}$ ho toh ($\sqrt{a} \cup \sqrt{b}$) se upar-neeche multiply - (a−b) formula se root gayab.	$1/(\sqrt{5}-2) \times (\sqrt{5}+2)/(\sqrt{5}+2) = \sqrt{5} + 2$
$a^m \times a^n = a^{m+n}$; $a^m/a^n = a^{m-n}$; $(a^m)^n = a^{mn}$	Multiply = powers add; divide = powers minus; power of power = powers multiply. Base SAME hona chahiye.	$3^5 \times 3^{-2} = 3^3 = 27$
$a^0 = 1$; $a^{-n} = 1/a^n$; $a^{1/n} = \sqrt[n]{a}$; $a^{m/n} = \sqrt[n]{(a^m)}$	Zero power = 1. Minus power = palat do. Fraction power = root. $64^{2/3}$ matlab cube root of 64, phir square.	$64^{2/3} = (4)^2 = 16$; $64^{-2/3} = 1/16$
Same base banao → exponents barabar rakh do	$8 = 2^3$, $125 = 5^3$, $1/81 = 3^{-4}$. Dono side same base aaye toh powers equate karo - equation solved.	$3^x = 1/81 = 3^{-4} \rightarrow x = -4$
rational ± irrational = irrational; irrational × non-zero rational = irrational	Ek irrational ke saath rational jodo ya guna karo - result irrational hi rahega (rational × rational ke alawa).	$3 + \sqrt{2}$, $\sqrt{2} \times 4$ dono irrational

3.2 Concept Checklist - Ye Sab Aana Hai

- **Irrational ka matlab:** decimal na terminate ho, na repeat. √2, π, 0.101001000... sab irrational.
- **0.999... = 1 exactly** - proof: x = 0.999..., 10x = 9.999..., 10x − x = 9 → x = 1. MCQ favourite!
- **Between any two rationals infinitely many irrationals** (aur vice versa) - density. Example banane ke liye surds use karo: √(a/b) forms.
- **√n irrational jab tak n perfect square na ho** - 2, 3, 5, 6, 7 sabke roots irrational; √4, √9, √16 rational.
- **Irrationality proof ka skeleton:** maan lo p/q (lowest terms) → square karo → prime p, q ko divide karta hai → p, q dono divide karte hain → contradiction. Iska flow ratt lo.
- **Compare surds:** sabko same root ke andar le aao - $(\sqrt{7})^2 = 7$, $(2\sqrt{2})^2 = 8$, $(3)^2 = 9$ → order mil gaya.
- **Conjugate pair magic:** x = 2 + √3 ho toh 1/x = 2 − √3 (multiply karke check karo). x ± 1/x ke questions isi se khul te hain.
- **Telescoping sum:** 1/(√2+1) + 1/(√3+√2) + ... har term rationalise karne par √(next) − √(prev) bachta hai - chain collapse ho jati hai.
- **Nested radical:** √(7 + 4√3) = √(4 + 3 + 2·2·√3) = 2 + √3. Pattern: m + n = given, mn = inner term.

3.3 Speed Tricks & Trap Alarms

Instant trick	Fire it when...
Terminating? REDUCE first, then check: denominator = $2^m \times 5^n$ only	"Without long division, state whether the decimal terminates".
Pure recurring 0.abab... = block/99...9. Mixed: (all digits - non-repeating)/99..900..0	Convert recurring decimals to p/q form.
Compare surds: same root index banao, then compare inside values	"Arrange in increasing order".
Rationalise: multiply by the conjugate. Telescoping sum collapses to (last - first)	Surds in denominators; 1/(√a+√b) chains.
Same base first: $8 = 2^3$, $125 = 5^3$, $1/81 = 3^{-4}$ → equate exponents	Every exponent equation.
Irrationality skeleton: assume p/q → square → prime lemma → contradiction	"Prove √2 / √3 / √5 is irrational".
rational ± irrational = irrational; irrational × non-zero rational = irrational	True/false closure statements.

3.4 Most Important Questions (10)

Ye wo questions hain jo **book ke exercises + exam pattern** se seedhe uthaye gaye hain - inka style, numbers aur twist exam mein aane ke sabse zyada chances hain. Solution chhota hai par poora logic covered hai.

IMP Q1 | Terminating test (T2)

Q1. Without performing long division, decide whether each decimal expansion terminates: (i) 17/8, (ii) 13/3125, (iii) 7/12, (iv) 64/455, (v) 6/15.

Step 1. Given: five fractions. Needed: terminating or not, via the $2^m \times 5^n$ test on the REDUCED denominator.

Step 2. (i) 17/8: $8 = 2^3 = 2^3 \times 5^0$. Only 2s, so it terminates (in fact 2.125).

Step 3. (ii) 13/3125: $3125 = 5^5$. Only 5s, so it terminates.

Step 4. (iii) 7/12: $12 = 2^2 \times 3$. The stray 3 can never be absorbed into a power of 10, so it recurs.

Step 5. (iv) 64/455: $455 = 5 \times 7 \times 13$. The primes 7 and 13 force a recurring decimal.

Step 6. (v) 6/15: reduce first! 6/15 = 2/5, and $5 = 5^1$, so it terminates (0.4). Testing $15 = 3 \times 5$ without reducing would give the wrong verdict.

Answer: (i) terminates (ii) terminates (iii) recurs (iv) recurs (v) terminates (6/15 = 2/5 = 0.4)

Hinglish: *Sabse pehla kaam: fraction ko lowest terms mein lao. Phir denominator ko prime factors mein todo. Sirf 2 aur 5 dikh rahe hain? Decimal ruk jayega. Koi aur prime (3, 7, 13...) dikh gaya? Chalti-firti decimal aayegi. (v) wala question isliye diya hai kyunki students 15 dekh ke turant 'recur' bol dete hain - pehle 3 cancel karo!*

IMP Q2 | Pure recurring to p/q (T3)

Q2. Express in p/q form: (i) 0.777... (the 7 repeating), (ii) 0.373737... (the block 37 repeating), (iii) 0.090909... (the block 09 repeating).

Step 1. Given: three pure recurring decimals (repeat starts immediately). Needed: exact fractions.

Step 2. (i) Let x = 0.777... . Multiply by 10 (one repeating digit → shift by one place): 10x = 7.777... .

Step 3. Subtract: 10x - x = 7.777... - 0.777... = 7. So 9x = 7 and x = 7/9.

Step 4. (ii) Let x = 0.373737... . Two repeating digits → multiply by 100: 100x = 37.3737... .

Step 5. Subtract: 100x - x = 37. So 99x = 37 and x = 37/99.

Step 6. (iii) Let x = 0.090909... . Two repeating digits → 100x = 9.0909... . Subtract: 99x = 9, x = 9/99 = 1/11.

Answer: (i) 7/9 (ii) 37/99 (iii) 1/11

Hinglish: *Rule simple hai: jitne digit repeat ho rahe hain, utne 9 denominator mein. Ek digit repeat = 9, do digit = 99. 0.777... = 7/9, 0.3737... = 37/99. Bas repeating block ke digits upar likh do aur neeche utne hi 9. (iii) mein 9/99 ko 1/11 tak reduce karna mat bhoolna.*

IMP Q3 | Surd simplification (T7)

Q3. Simplify into k√( ) form: (i) √72, (ii) √245, (iii) √175, (iv) √(147).

Step 1. Given: four radicals. Needed: extract every perfect-square factor.

Step 2. (i) $72 = 36 \times 2$, so $\sqrt{72} = \sqrt{36} \cdot \sqrt{2} = 6\sqrt{2}$.

Step 3. (ii) $245 = 49 \times 5$, so $\sqrt{245} = 7\sqrt{5}$.

Step 4. (iii) $175 = 25 \times 7$, so $\sqrt{175} = 5\sqrt{7}$.

Step 5. (iv) $147 = 49 \times 3$, so $\sqrt{147} = 7\sqrt{3}$.

Answer: (i) 6√2 (ii) 7√5 (iii) 5√7 (iv) 7√3

Hinglish: *Andar sabse bada perfect square dhoondo aur bahar nikalo. 72 = 36×2, 36 ka root 6, toh 6√2. Yaad rakho: 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144 - ye aapke dost hain. Jitna bada square nikaloge, utna chhota root; 72 ko 4×18 karke √18 ko phir simplify karna double kaam hai.*

IMP Q4 | Irrationality proof (T5)

Q4. Prove that √3 is irrational.

Step 1. Given: a claim to contradict. Assume $\sqrt{3} = p/q$ where p, q are integers, $q \neq 0$, and the fraction is in lowest terms (p and q share no common prime).

Step 2. Squaring both sides: $3 = p^2/q^2$, so $p^2 = 3q^2$.

Step 3. Then 3 divides p^2. Since 3 is prime, 3 divides p itself (a prime dividing a square must divide the number). Write $p = 3r$.

Step 4. Substitute: $(3r)^2 = 3q^2 \Rightarrow 9r^2 = 3q^2 \Rightarrow q^2 = 3r^2$.

Step 5. So 3 divides q^2, hence 3 divides q.

Step 6. Now 3 divides both p and q - contradicting lowest terms. The assumption is false, therefore √3 is irrational. ∎

Answer: √3 is irrational (by contradiction; 3 divides both p and q, contradicting lowest terms).

Hinglish: *Proof ka 4-step skeleton yaad karo: (1) assume p/q lowest terms, (2) square karo - prime upar aa jata hai, (3) prime dono (p aur q) ko divide karta nikla, (4) lowest terms se contradiction - done! √2 mein prime 2 tha, yahan 3 hai, √5 mein 5 hoga. Bas prime badlo, skeleton same.*

IMP Q5 | Rationalising a + b√c denominator (T9)

Q5. Rationalise the denominator and simplify: $1/(3 + 2\sqrt{2})$.

Step 1. Given: denominator $3 + 2\sqrt{2}$. Its conjugate is $3 - 2\sqrt{2}$ (flip the sign of the surd term).

Step 2. Multiply numerator and denominator by the conjugate:

Step 3. $= (3 - 2\sqrt{2})/((3 + 2\sqrt{2})(3 - 2\sqrt{2}))$.

Step 4. Denominator: $3^2 - (2\sqrt{2})^2 = 9 - 8 = 1$ (difference of squares).

Step 5. So the expression equals $3 - 2\sqrt{2}$. Check numerically: $1/(3 + 2.828...) = 1/5.828... \approx 0.1716$, and $3 - 2.828... = 0.1716$ ✓.

Answer: $3 - 2\sqrt{2}$

Hinglish: *Conjugate ka matlab: sirf beech ka sign ulta. $3 + 2\sqrt{2}$ ka conjugate $3 - 2\sqrt{2}$. Multiply karo, denominator $(a+b)(a-b) = a^2-b^2$ ban jata hai. Yahan 9-8=1 nikla - jackpot! Answer $3-2\sqrt{2}$. Aise lucky cases (denominator 1) exam mein bahut aate hain, isliye denominator ko poori tarah simplify karna kabhi mat chhodna.*

IMP Q6 | Comparing surds (T8)

Q6. Arrange in increasing order: $\sqrt{7}$, $2\sqrt{2}$, $\sqrt{5}$, $3\sqrt{3}/2$.

Step 1. Given: four positive surds. All are positive, so squaring preserves the order.

Step 2. $\sqrt{7} \to 7$. $2\sqrt{2} \to (2\sqrt{2})^2 = 4 \times 2 = 8$. $\sqrt{5} \to 5$. $(3\sqrt{3})/2 \to (9 \times 3)/4 = 27/4 = 6.75$.

Step 3. Order of the squares: $5 < 6.75 < 7 < 8$.

Step 4. So the original order: $\sqrt{5} < 3\sqrt{3}/2 < \sqrt{7} < 2\sqrt{2}$.

Step 5. Check: $\sqrt{5} \approx 2.236$, $3\sqrt{3}/2 \approx 2.598$, $\sqrt{7} \approx 2.646$, $2\sqrt{2} \approx 2.828$ ✓.

Answer: $\sqrt{5} < 3\sqrt{3}/2 < \sqrt{7} < 2\sqrt{2}$

Hinglish: *Compare karne ka best trick: dono ko square karo. $2\sqrt{2}$ ka square $(2)^2 \times 2 = 8$ hota hai - coefficient bhi square hota hai, ye bhoolna sabse common galti hai! $(3\sqrt{3}/2)^2 = 9 \times 3/4 = 6.75$. Ab squares ko order karo aur original order likho. Numerical check se confirm kar lo.*

IMP Q7 | Same-base equations (T11)

Q7. Solve: (i) $3^x = 1/81$, (ii) $5^{(x-2)} = 125$, (iii) $(2/3)^x = 27/8$.

Step 1. Given: three exponential equations. Strategy: one base on both sides, then equate exponents.

Step 2. (i) $1/81 = 1/3^4 = 3^{-4}$. So $3^x = 3^{-4} \Rightarrow x = -4$.

Step 3. (ii) $125 = 5^3$. So $5^{(x-2)} = 5^3 \Rightarrow x - 2 = 3 \Rightarrow x = 5$.

Step 4. (iii) $27/8 = 3^3/2^3 = (3/2)^3 = (2/3)^{-3}$. So $x = -3$.

Step 5. Check: $(2/3)^{-3} = (3/2)^3 = 27/8$ ✓.

Answer: (i) x = -4 (ii) x = 5 (iii) x = -3

Hinglish: *Exponent equation mein dono taraf SAME base lao. $1/81$ ko 3^{-4} dekho - negative exponent ka perfect use. $(2/3)^x = 27/8$ mein ulta fraction banana padta hai: $27/8 = (3/2)^3 = (2/3)^{-3}$. Fraction ka negative exponent = fraction flip. Ye teen patterns exam ke favourites hain.*

IMP Q8 | Fractional and negative exponents (T10)

Q8. Evaluate without a calculator: (i) $(81/16)^{-34y}$... rewrite: (81/16)^(-3/4), (ii) 8^(2/3) + 4^(-1/2), (iii) $(5^2/\sqrt{5})^2$.

Step 1. Given: expressions with fractional and negative exponents.

Step 2. (i) (81/16)^(-3/4): the negative exponent flips the fraction; the 1/4 takes a fourth root. (16/81)^(3/4) = $(\sqrt[4]{(16/81)})^3 = (2/3)^3 = 8/27$.

Step 3. (ii) 8^(2/3) = $(\sqrt[3]{8})^2 = 2^2 = 4$. 4^(-1/2) = $1/\sqrt{4} = 1/2$. Sum = 4 + 1/2 = 9/2.

Step 4. (iii) $5^2/\sqrt{5} = 5^2/5$^(1/2) = 5^(3/2). Squaring: $5^3 = 125$.

Step 5. Check (iii) numerically: 25/2.236 ≈ 11.18; $11.18^2 \approx 125$ ✓.

Answer: (i) 8/27 (ii) 9/2 (iii) 125

Hinglish: *Fractional exponent do kaam karta hai: neeche wala number ROOT, upar wala POWER. 8^(2/3): cube root of 8 = 2, phir square = 4. Negative exponent: flip karo. Root pehle lo, power baad mein - numbers chhote rehte hain. (i) mein (81/16) ko flip karke 16/81 banana zaroori tha.*

IMP Q9 | Conjugate pairs x and 1/x (T14)

Q9. If $x = 2 + \sqrt{3}$, find the value of $x^2 + 1/x^2$ without evaluating x numerically beyond the given form.

Step 1. First find 1/x by rationalising: $1/(2 + \sqrt{3}) = (2 - \sqrt{3})/((2 + \sqrt{3})(2 - \sqrt{3})) = (2 - \sqrt{3})/(4 - 3) = 2 - \sqrt{3}$.

Step 2. So x and 1/x are conjugates: $x = 2 + \sqrt{3}$, $1/x = 2 - \sqrt{3}$.

Step 3. Square both: $x^2 = 4 + 4\sqrt{3} + 3 = 7 + 4\sqrt{3}$; $(1/x)^2 = 7 - 4\sqrt{3}$.

Step 4. Add: $x^2 + 1/x^2 = 14$ (the surd terms cancel).

Step 5. Check numerically: x ≈ 3.732, $x^2 \approx 13.93$, $1/x^2 \approx 0.0718$, sum ≈ 14.0 ✓.

Answer: 14

Hinglish: *Ye conjugate-pair trick poore algebra mein chalega. Pehle 1/x nikaalo - denominator 1 aaya, matlab x aur 1/x perfect conjugates. Square karke jodo: $\pm 4\sqrt{3}$ terms aapas mein uda do. Answer 14. Aage ke liye yaad rakho: $x^3 + 1/x^3 = (x + 1/x)^3 - 3(x + 1/x)$.*

IMP Q10 | Telescoping surd sum (T14)

Q10. Evaluate: $1/(\sqrt{2} + 1) + 1/(\sqrt{3} + \sqrt{2}) + 1/(2 + \sqrt{3})$.

Step 1. Key identity: $1/(\sqrt{(n+1)} + \sqrt{n}) = \sqrt{(n+1)} - \sqrt{n}$ (multiply by the conjugate; denominator becomes n+1-n = 1).

Step 2. First term: $1/(\sqrt{2} + 1) = \sqrt{2} - 1$.

Step 3. Second: $1/(\sqrt{3} + \sqrt{2}) = \sqrt{3} - \sqrt{2}$.

Step 4. Third: $1/(2 + \sqrt{3}) = \sqrt{4} - \sqrt{3} = 2 - \sqrt{3}$.

Step 5. Sum: $(\sqrt{2} - 1) + (\sqrt{3} - \sqrt{2}) + (2 - \sqrt{3}) = 2 - 1 = 1$ (everything in between cancels - telescoping).

Answer: 1

> ***Hinglish:*** *Ye 'telescoping' pattern dekho: har term rationalise karne par root ka difference banta hai: √2-1, √3-√2, 2-√3. Jodne par beech ke saare roots gayab, sirf first ka minus aur last ka plus bachta hai: 2 - 1 = 1. Agar 1/(√(n+1)+√n) type dikhe, turant conjugate socho.*

3.5 Rapid-Fire MCQs (14)

Rapid fire: option padho, dimaag mein answer banao, phir Answer line se check karo. Galat hua toh Answer line ka logic padho - wahi 1 line exam mein marks bachayegi.

1. Which of the following is irrational?

(A) 0.101001000100001... **(B)** 0.25 **(C)** 3/8 **(D)** √49

Ans: (A) · The gaps between 1s keep growing, so no block repeats: non-terminating, non-recurring, hence irrational. The others terminate or simplify (0.25, 3/8, √49 = 7).

2. 0.333... (the 3 repeating) written as a fraction is:

(A) 1/3 **(B)** 3/10 **(C)** 33/100 **(D)** 1/30

Ans: (A) · One repeating digit: x = 3/9 = 1/3. Check: 1/3 = 0.333... ✓.

3. Without division, the decimal expansion of 77/210 is:

(A) terminating **(B)** non-terminating recurring **(C)** non-terminating non-recurring **(D)** cannot be decided

Ans: (B) · 77/210 = 11/30 (cancel 7). 30 = 2 × 3 × 5 has the prime 3, so it recurs.

4. Which statement is TRUE?

(A) Z ⊂ N **(B)** Q ⊂ Z **(C)** N ⊂ Q **(D)** R ⊂ Q

Ans: (C) · Naturals sit inside integers, integers inside rationals, rationals inside reals: N ⊂ Z ⊂ Q ⊂ R. So N ⊂ Q is true.

5. 0.4777... (the 4 not repeating, the 7 repeating) equals:

(A) 43/90 **(B)** 47/100 **(C)** 47/90 **(D)** 43/99

Ans: (A) · (47 - 4)/90 = 43/90: one 9 for the repeating digit, one 0 for the fixed digit.

6. √2 × √8 equals:

(A) √10 **(B)** 4 **(C)** 2√4 **(D)** 16

Ans: (B) · √2 × √8 = √16 = 4. Roots multiply by multiplying radicands.

7. Which is the largest?

(A) 2√3 **(B)** 3√2 **(C)** 4 **(D)** √15

Ans: (B) · Squares: 12, 18, 16, 15. The largest square is 18, so 3√2 is largest.

8. If $3^x = 1/243$, then x =

(A) 5 **(B)** -5 **(C)** 1/5 **(D)** -243

Ans: (B) · $1/243 = 1/3^5 = 3^{-5}$, so x = -5.

9. $1/(\sqrt{5} - 2)$ equals:

(A) $\sqrt{5} + 2$ **(B)** $\sqrt{5} - 2$ **(C)** $(\sqrt{5}+2)/3$ **(D)** 1/3

Ans: (A) · Multiply by $(\sqrt{5} + 2)$: denominator $(\sqrt{5})^2 - 2^2 = 5 - 4 = 1$. So the value is $\sqrt{5} + 2$.

10. $(\sqrt{7} + \sqrt{5})(\sqrt{7} - \sqrt{5})$ equals:

(A) 2 **(B)** 12 **(C)** $\sqrt{2}$ **(D)** 35

Ans: (A) · $(\sqrt{7})^2 - (\sqrt{5})^2 = 7 - 5 = 2$.

11. If $x = 3 + 2\sqrt{2}$, then 1/x equals:

(A) $3 - 2\sqrt{2}$ **(B)** $2\sqrt{2} - 3$ **(C)** $1/(3+2\sqrt{2})$ simplified to 1/3 **(D)** 6

Ans: (A) · $1/(3+2\sqrt{2}) = (3-2\sqrt{2})/((3)^2-(2\sqrt{2})^2) = (3-2\sqrt{2})/(9-8) = 3 - 2\sqrt{2}$.

12. (16/81)^(-1/4) equals:

(A) 3/2 **(B)** 2/3 **(C)** -3/2 **(D)** 81/16

Ans: (A) · Negative exponent flips: $(81/16)^{(1/4)} = \sqrt[4]{81} / \sqrt[4]{16} = 3/2$.

13. $(\sqrt{3} + \sqrt{2})^2$ equals:

(A) $5 + 2\sqrt{6}$ **(B)** $5 + \sqrt{6}$ **(C)** 13 **(D)** 5

Ans: (A) · $3 + 2\sqrt{6} + 2 = 5 + 2\sqrt{6}$. The cross term $2\sqrt{3}\sqrt{2} = 2\sqrt{6}$ is essential.

14. $1/(\sqrt{9} - \sqrt{8})$ equals:

(A) $3 + 2\sqrt{2}$ **(B)** $3 - 2\sqrt{2}$ **(C)** 1 **(D)** $3 + \sqrt{8}$

Ans: (A) · $\sqrt{9} = 3$, $\sqrt{8} = 2\sqrt{2}$. Multiply by $(3 + 2\sqrt{2})$: denominator 9 - 8 = 1, value $3 + 2\sqrt{2}$.

Chapter 4: Algebraic Identities and Factorisation ★ MOST IMPORTANT

> **Ek line mein chapter**
> Pura chapter ek toolbox hai - 8 identities + 5 factorisation moves. Exam mein sabse zyada marks yahin se: near-number evaluation, x+1/x chains, cube bridges aur zero-sum shortcut. Identities ko "kab use karna hai" ke saath yaad karo, akela formula kaam nahi karta.

4.1 Saare Formulas - Easy Bhasha Mein

Har formula ke saath uska **easy matlab** aur ek **tiny example** hai - formula akela kabhi mat rattro, example ke saath yaad karo. Exam mein pehle 2 second sochho kaunsa formula fire karna hai, phir likho.

Formula (ratt lo)	Easy bhasha mein matlab	Tiny example
$(a + b)^2 = a^2 + 2ab + b^2$	Square + DOUBLE product + square. Beech wala 2ab hi asli identity hai - wo bhool gaye toh sirf squares likhna galat.	$(2x + 3)^2 = 4x^2 + 12x + 9$
$(a - b)^2 = a^2 - 2ab + b^2$	Sirf MIDDLE term minus hota hai, last term plus rehta hai. $a^2 - 2ab - b^2$ likhna classic disaster hai.	$(3x - 4y)^2 = 9x^2 - 24xy + 16y^2$
$a^2 - b^2 = (a + b)(a - b)$	Do direction mein chalta hai: factorise karo YA quick multiply karo. "Without direct multiplication" dikhe toh yehi.	$51 \times 49 = 50^2 - 1^2 = 2499$
$(x + a)(x + b) = x^2 + (a+b)x + ab$	x ka coefficient = dono numbers ka SUM, constant = PRODUCT. Sum-product yaad rakhna, expansion nahi karna.	$(x + 4)(x + 7) = x^2 + 11x + 28$
$(a + b + c)^2 = a^2+b^2+c^2 + 2ab + 2bc + 2ca$	Teen squares + har JODI ka double (3 jodiyan hote hain, bhoolna mat). Cross term ka sign = uske dono parents ke signs ka product.	$(2p + 3q - 4r)^2$ mein 12pq, −24qr, −16pr
$(a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$	Signs same order mein: $(a+b)^3$ sab plus; $(a-b)^3$ mein + − + − pattern (last term minus!).	$103^3 = 1000000 + 90000 + 2700 + 27 = 1092727$
$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$ $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$	Bracket apna sign rakhta hai; trinomial ke beech wale term ka sign OPPOSITE. $8a^3+27b^3 = (2a)^3+(3b)^3$ pehle banao.	$x^3 - 27 = (x-3)(x^2+3x+9)$
Bridges: $a^2+b^2 = (a+b)^2 - 2ab$ $a^3+b^3 = (a+b)^3 - 3ab(a+b)$	a+b aur ab diye ho toh ye bridges DIRECT answer dete hain. Given values kabhi random nahi hote - bridge laga do.	$a+b=12$, $ab=27 \rightarrow a^3+b^3 = 1728 - 972 = 756$
Crown: $a^3+b^3+c^3 - 3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$	Agar **$a+b+c = 0$** ho toh RIGHT side zero → **$a^3+b^3+c^3 = 3abc$**. Ye shortcut exam ka king hai.	$12^3 + 1^3 + (-13)^3 = 3 \times 12 \times 1 \times (-13) = -468$

Formula (ratt lo)	Easy bhasha mein matlab	Tiny example
$x + 1/x = s \rightarrow x^2 + 1/x^2 = s^2 - 2 \rightarrow x^3 + 1/x^3 = s^3 - 3s$	Chain chalao: har baar square/cube karo aur 2 (ya 3s) minus. $x - 1/x = d$ ho toh PLUS 2 / PLUS 3d hota hai.	$s = 3 \rightarrow 7 \rightarrow 18 \rightarrow x^4+1/x^4 = 47$
$(a+b)^2 - (a-b)^2 = 4ab$; $(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$	Jodi wale identities: difference = 4ab (quick multiply engine), sum = $2(a^2+b^2)$.	$5.5^2 - 4.5^2 = 4 \times 5.5 \times 4.5 = 99$
Middle-term split: $x^2 + px + q \rightarrow$ 2 numbers jinka sum p, product q	$x^2+7x+12$: 4 aur 3 (sum 7, product 12) $\rightarrow (x+4)(x+3)$. General ax^2+bx+c mein product = a×c lo.	$x^2 - 11x - 26 = (x-13)(x+2)$

4.2 Concept Checklist - Ye Sab Aana Hai

- **Golden order of factorisation:** (1) HCF common → (2) identity → (3) grouping → (4) middle-term split → (5) substitution. HCF SABSE PEHLE, hamesha.
- **Near numbers:** $98^2 = (100-2)^2$, $103^3 = (100+3)^3$, $999^2 + 2 \cdot 999 + 1 = (999+1)^2 = 1000000$. Round number ke (round ± chhota) mein todo.
- **Zero-sum instant:** a+b+c = 0 dikha → cubes ka sum = 3abc likh ke aage badho. $25^3 - 75^3 + 50^3 = 3 \times 25 \times (-75) \times 50$.
- **Crown ka converse:** $a^3+b^3+c^3 = 3abc$ ho toh either a+b+c = 0 YA a = b = c.
- **$x^4 + x^2 + 1 = (x^2+x+1)(x^2-x+1)$** - add-subtract x^2 trick. $x^4 - 81$ jaise chains mein har layer ko todte raho.
- **Cubic factoring:** x = 1, −1, 2, −2 try karo root ke liye; root mil gaya toh (x − root) se divide.
- **Sign of cross terms in $(a+b+c)^2$:** har cross term ka sign uske dono parents ke signs ka product. Count karo - exactly 3 pairs, har ek twice.
- **Cube bridge scaled version:** 2x+3y = 13, xy = 6 → $8x^3+27y^3 = (2x+3y)^3 - 3(2x)(3y)(2x+3y)$. Pehle block banao, phir bridge.
- **Verify karna mana nahi:** factorisation ke baad expand karke check karo - 30 second mein galat answer pakda jata hai.

4.3 Speed Tricks & Trap Alarms

Instant trick	Fire it when...
Near round numbers: $51 \times 49 = 50^2 - 1$; $102^2 = 100^2 + 2 \cdot 100 \cdot 2 + 4$	"Evaluate without direct multiplication".
Given a + b and ab: $a^2 + b^2 = (a+b)^2 - 2ab$; $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$	Bridge questions - the two givens are never random.
$a + b + c = 0 \rightarrow a^3 + b^3 + c^3 = 3abc$, instantly	Zero-sum cubes like $25^3 - 75^3 + 50^3$.
$x + 1/x = s$ → square it: $x^2 + 1/x^2 = s^2 - 2$ → keep chaining up	The whole x + 1/x family.
Spot perfect cubes (8, 27, 64, 125, a^3b^3) → apply $a^3 \pm b^3$	$27a^3 + 64$ type expressions.

Instant trick	Fire it when...
Cubic factoring: test x = 1, -1, 2, -2 for a root, then divide	$x^3 - 3x^2 - 9x - 5$ type.
HCF first, ALWAYS - before any identity or grouping	Every factorisation: move 1 of the golden sequence.

4.4 Most Important Questions (12)

Ye wo questions hain jo **book ke exercises + exam pattern** se seedhe uthaye gaye hain - inka style, numbers aur twist exam mein aane ke sabse zyada chances hain. Solution chhota hai par poora logic covered hai.

IMP Q1 | Near-number squares (T3)

Q1. Evaluate using identities: (i) 102^2, (ii) 98^2, (iii) 9.7^2.

Step 1. Given: three squares near round numbers. Write each as (100 ± k) or (10 ± k).

Step 2. (i) $102^2 = (100+2)^2 = 10000 + 400 + 4 = 10404$.

Step 3. (ii) $98^2 = (100-2)^2 = 10000 - 400 + 4 = 9604$.

Step 4. (iii) $9.7^2 = (10 - 0.3)^2 = 100 - 6 + 0.09 = 94.09$.

Answer: (i) 10404 (ii) 9604 (iii) 94.09

Hinglish: *Round number ke paas wale squares ka JUGAAD: $(100\pm2)^2$. $98^2 = 10000 - 400 + 4 = 9604$. Dhyan do: 9.7^2 mein cross term $2\times10\times0.3 = 6$, aur $0.3^2 = 0.09$ - decimals ka square chhota hota hai, 0.9 nahi!*

IMP Q2 | Square-sum identities (T1/T11)

Q2. If a + b = 10 and a - b = 4, find (i) $a^2 + b^2$ and (ii) ab - all in one line each, using identities.

Step 1. Given: the pair-sum and pair-difference. Needed: sum of squares and the product.

Step 2. Add the two squares identities: $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$.

Step 3. So $a^2 + b^2 = (10^2 + 4^2)/2 = (100 + 16)/2 = 58$.

Step 4. Subtract: $(a+b)^2 - (a-b)^2 = 4ab$.

Step 5. So $ab = (100 - 16)/4 = 84/4 = 21$.

Step 6. Check: $(a+b)(a-b) = 10\times4 = 40 = a^2 - b^2$; with $a^2+b^2 = 58$ we get $a^2 = 49$, $b^2 = 9$, a = 7, b = 3 ✓ (ab = 21 ✓).

Answer: $a^2 + b^2 = 58$, ab = 21

Hinglish: *Ye identity-pair exam ka shortcut bomb hai: $(a+b)^2+(a-b)^2 = 2(a^2+b^2)$ aur $(a+b)^2-(a-b)^2 = 4ab$. Dono values plug karo - 58 aur 21. Solving for a, b optional hai, sirf check ke liye.*

IMP Q3 | Middle-term splitting (T6)

Q3. Factorise: (i) $x^2 + 7x + 12$, (ii) $x^2 - 5x - 36$, (iii) $x^2 - 11x + 24$.

Step 1. Given: three monic quadratics. Find two numbers with the right SUM and PRODUCT.

Step 2. (i) sum 7, product 12 → 3 and 4: $(x + 3)(x + 4)$.

Step 3. (ii) sum -5, product -36 → -9 and +4: $(x - 9)(x + 4)$.

Step 4. (iii) sum -11, product 24 → -8 and -3: $(x - 8)(x - 3)$.

Step 5. Checks: 3+4 = 7, 3×4 = 12 ✓; -9+4 = -5, -9×4 = -36 ✓; -8-3 = -11, 24 ✓.

Answer: (i) (x+3)(x+4) (ii) (x-9)(x+4) (iii) (x-8)(x-3)

Hinglish: *Do numbers dhoondo: jiska sum middle coefficient ho, product last term ho. Sign logic: product positive toh dono same sign (sum ke hisaab se), product negative toh alag-alag. (ii) mein product -36 aur sum -5: -9 aur +4 jeette hain.*

IMP Q4 | x + 1/x chain (T10)

Q4. If $x + 1/x = 5$, find (i) $x^2 + 1/x^2$ and (ii) $x^4 + 1/x^4$.

Step 1. Given: $x + 1/x = 5$. Square both sides: $x^2 + 2 + 1/x^2 = 25$.

Step 2. (i) $x^2 + 1/x^2 = 25 - 2 = 23$.

Step 3. Square again: $(x^2 + 1/x^2)^2 = x^4 + 2 + 1/x^4 = 23^2 = 529$.

Step 4. (ii) $x^4 + 1/x^4 = 529 - 2 = 527$.

Answer: (i) 23 (ii) 527

Hinglish: *Chain method: square karo, 2 minus karo, phir dobara square karo, phir 2 minus. 5 → 23 → 527. Ek hi line mein: $(x+1/x)^2 - 2 = x^2+1/x^2$. Ye chain chalti hi jati hai - x^6 ke liye x^3 wale bridge ka use hota hai.*

IMP Q5 | Cube sum bridge (T11)

Q5. If $a + b = 12$ and $ab = 27$, find $a^3 + b^3$ and $a^2 + b^2$.

Step 1. Given: pair-sum and product. Cube bridge: $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$.

Step 2. $a^3 + b^3 = 12^3 - 3(27)(12) = 1728 - 972 = 756$.

Step 3. Square identity: $a^2 + b^2 = (a+b)^2 - 2ab = 144 - 54 = 90$.

Answer: $a^3 + b^3 = 756$, $a^2 + b^2 = 90$

Hinglish: *Symmetric quantities ka rule: sab kuch (sum, product) se nikhlo. Cube bridge: $(a+b)^3 - 3ab(a+b)$. 1728 - 972 = 756. Square: 144 - 54 = 90. Solving a, b se bhi hoga but identity wala path 2 line ka hai.*

IMP Q6 | Crown identity evaluation (T9)

Q6. Using a suitable identity, evaluate $12^3 + 1^3 + (-13)^3$.

Step 1. Spot the zero sum: $12 + 1 + (-13) = 0$.

Step 2. Zero-sum shortcut: if $a + b + c = 0$ then $a^3 + b^3 + c^3 = 3abc$.

Step 3. So the value = $3 \times 12 \times 1 \times (-13) = -468$.

Answer: -468

Hinglish: *Numbers dekho: 12, 1, -13 - sum zero! Turant shortcut: cubes ka sum = 3abc = -468. Bina koi cube nikaale. Ye 'zero-sum spotting' exam mein 3 second ka kaam hai - 5 marks bachate hain.*

IMP Q7 | Cubic factorisation by root (T12)

Q7. Factorise completely: $x^3 - 3x^2 - 9x - 5$.

Step 1. Hunt for an integer root: try $x = -1$: $(-1)^3 - 3(1) - 9(-1) - 5 = -1 - 3 + 9 - 5 = 0$ ✓. So $(x + 1)$ is a factor.

Step 2. Divide: $x^3 - 3x^2 - 9x - 5 = (x + 1)(x^2 - 4x - 5)$.

Step 3. (Verify by expanding: $x^3 - 4x^2 - 5x + x^2 - 4x - 5 = x^3 - 3x^2 - 9x - 5$ ✓.)

Step 4. Factorise the quadratic: $x^2 - 4x - 5$: sum -4, product -5 → -5 and +1: $(x - 5)(x + 1)$.

Step 5. Final: $(x + 1)^2(x - 5)$.

Answer: $(x + 1)^2(x - 5)$

Hinglish: *Cubic factorisation ka opening: chhote integers try karo. x = -1 par zero mila, toh (x+1) factor. Long division ya coefficient comparison se quadratic mila: x^2-4x-5, wo phir middle-term split. Final $(x+1)^2(x-5)$.*

IMP Q8 | Crown identity forward use (T9)

Q8. If $x + y + z = 6$ and $xy + yz + zx = 11$, find $x^3 + y^3 + z^3 - 3xyz$.

Step 1. First get the sum of squares: $x^2+y^2+z^2 = (x+y+z)^2 - 2(xy+yz+zx) = 36 - 22 = 14$.

Step 2. Crown identity: $x^3+y^3+z^3-3xyz = (x+y+z)(x^2+y^2+z^2 - xy - yz - zx)$.

Step 3. The bracket: $14 - 11 = 3$.

Step 4. Value = $6 \times 3 = 18$.

Answer: 18

Hinglish: *Do steps: pehle $x^2+y^2+z^2 = 36-22 = 14$. Phir crown identity: bracket = 14-11 = 3. Multiply: 6×3 = 18. Zero-sum nahi tha isliye poora identity use hua.*

IMP Q9 | Scaled square bridge (T11)

Q9. If $9x^2 + 25y^2 = 181$ and $xy = 2$, find $3x - 5y$.

Step 1. Note $9x^2 = (3x)^2$ and $25y^2 = (5y)^2$.

Step 2. $(3x - 5y)^2 = 9x^2 - 2(3x)(5y) + 25y^2 = 181 - 30xy = 181 - 60 = 121$.

Step 3. So $3x - 5y = \pm 11$.

Answer: $3x - 5y = 11$ or -11

Hinglish: *$9x^2$ aur $25y^2$ ko $(3x)^2$, $(5y)^2$ dekho. Cross term $2\times3x\times5y = 30xy = 60$. $181 - 60 = 121 = 11^2$. Sign: dono values possible - ±11. Ye 'scaled variables' wala pattern har identity mein chalta hai.*

IMP Q10 | Zero-sum with big numbers (T9)

Q10. Without cubing any number, evaluate $25^3 - 75^3 + 50^3$.

Step 1. Sum check: $25 - 75 + 50 = 0$. So $a = 25$, $b = -75$, $c = 50$ has $a + b + c = 0$.

Step 2. Zero-sum shortcut: $a^3 + b^3 + c^3 = 3abc$.

Step 3. Value $= 3 \times 25 \times (-75) \times 50$.

Step 4. $25 \times 75 = 1875$; $1875 \times 50 = 93{,}750$; times 3 = 281,250. With the minus sign: -281,250.

Answer: -281,250

Hinglish: *$25 - 75 + 50 = 0$ - zero-sum! -75 ko b banake shortcut lagao: $3\times25\times(-75)\times50 = -281250$. Teen cubes (15625 + ... + 125000) karne ki zaroorat hi nahi. Sign: ek negative factor, toh result negative.*

IMP Q11 | Decimal cube sum (T3/T9)

Q11. Without cubing, evaluate $0.86^3 + 0.14^3$.

Step 1. $a + b = 0.86 + 0.14 = 1$.

Step 2. Bridge: $a^3 + b^3 = (a+b)^3 - 3ab(a+b) = 1 - 3ab$.

Step 3. $ab = 0.86 \times 0.14 = 0.1204$.

Step 4. Value $= 1 - 3(0.1204) = 1 - 0.3612 = 0.6388$.

Answer: 0.6388

Hinglish: *$0.86 + 0.14 = 1$ - yahi design hai. Bridge: $1 - 3(0.1204) = 0.6388$. Decimals ka product dhyan se: $86\times14 = 1204$, four decimal places. Cubes bilkul nahi nikaale.*

IMP Q12 | Bridge reversed for product (T11)

Q12. If $p^3 + q^3 = 35$ and $p + q = 5$, find pq.

Step 1. Bridge: $p^3 + q^3 = (p+q)^3 - 3pq(p+q)$.

Step 2. $35 = 125 - 3pq \cdot 5 = 125 - 15pq$.

Step 3. $15pq = 90$, so $pq = 6$.

Answer: pq = 6

Hinglish: *Bridge ka reverse use: $35 = 125 - 15pq$. $pq = 6$. Check: p, q roots of $t^2-5t+6 = (t-2)(t-3)$: $p=2$, $q=3 \rightarrow 8+27 = 35$ ✓.*

4.5 Rapid-Fire MCQs (15)

Rapid fire: option padho, dimaag mein answer banao, phir Answer line se check karo. Galat hua toh Answer line ka logic padho - wahi 1 line exam mein marks bachayegi.

1. 101^2 equals:

(A) 10201 **(B)** 10021 **(C)** 1021 **(D)** 10404

Ans: (A) · $(100+1)^2 = 10000 + 200 + 1 = 10201$.

2. 51×49 equals:

(A) 2499 **(B)** 2501 **(C)** 2490 **(D)** 2489

Ans: (A) · $(50+1)(50-1) = 2500 - 1 = 2499$.

3. $(a+b)^2 - (a-b)^2$ equals:

(A) 2ab **(B)** 4ab **(C)** $2(a^2+b^2)$ **(D)** 0

Ans: (B) · Subtracting the expansions cancels a^2 and b^2, leaving 4ab.

4. $x^2 + 7x + 12$ factorises as:

(A) (x+3)(x+4) **(B)** (x+2)(x+6) **(C)** (x+1)(x+12) **(D)** (x-3)(x-4)

Ans: (A) · $3 + 4 = 7$ and $3 \times 4 = 12$.

5. $9x^2 + 12x + 4$ equals:

(A) $(3x+2)^2$ **(B)** $(3x+4)^2$ **(C)** $(9x+2)^2$ **(D)** (3x+2)(3x-2)

Ans: (A) · $(3x)^2 + 2(3x)(2) + 2^2 = (3x+2)^2$.

6. If $x + 1/x = 2$, then $x^2 + 1/x^2$ equals:

(A) 2 **(B)** 4 **(C)** 6 **(D)** 0

Ans: (A) · $(x+1/x)^2 = x^2+2+1/x^2 = 4 \Rightarrow x^2+1/x^2 = 2$.

7. $(a + b + c)^2$ equals:

(A) $a^2+b^2+c^2+2ab+2bc+2ca$ **(B)** $a^2+b^2+c^2+ab+bc+ca$ **(C)** $a^2+b^2+c^2$ **(D)** $a^2+b^2+c^2+2abc$

Ans: (A) · Three squares plus TWICE each of the three pairwise products.

8. If $x + 1/x = 3$, then $x^3 + 1/x^3$ equals:

(A) 18 **(B)** 27 **(C)** 21 **(D)** 30

Ans: (A) · $x^3+1/x^3 = 27 - 3\cdot3 = 27 - 9 = 18$.

9. If $a + b + c = 0$, then $a^3 + b^3 + c^3$ equals:

(A) 3abc **(B)** 0 **(C)** -3abc **(D)** abc

Ans: (A) · Crown identity with a+b+c = 0 gives $a^3+b^3+c^3 = 3abc$.

10. The last three digits of 999^3 are:

(A) 299 **(B)** 001 **(C)** 999 **(D)** 997

Ans: (A) · $999^3 = (1000-1)^3 = 997002999$; last three digits 299.

11. $a^3 + b^3$ equals:

(A) $(a+b)(a^2-ab+b^2)$ **(B)** $(a+b)(a^2+ab+b^2)$ **(C)** $(a+b)^3$ **(D)** $(a-b)(a^2+ab+b^2)$

Ans: (A) · Standard factorisation; the trinomial's middle term is -ab.

12. If $a + b + c = 9$ and $ab + bc + ca = 26$, then $a^2 + b^2 + c^2$ equals:

(A) 29 **(B)** 55 **(C)** 107 **(D)** 29±... 33

Ans: (A) · $81 = a^2+b^2+c^2 + 52$, so the sum of squares is 29.

13. If $x + y = 5$ and $xy = 6$, then $x^3 + y^3$ equals:

(A) 35 **(B)** 125 **(C)** 53 **(D)** 63

Ans: (A) · $125 - 3\cdot6\cdot5 = 125 - 90 = 35$.

14. If $a^3 + b^3 = 152$ and $a + b = 8$, then ab equals:

(A) 15 **(B)** 12 **(C)** 19 **(D)** 16

Ans: (A) · $152 = 512 - 3ab(8) \Rightarrow 24ab = 360 \Rightarrow ab = 15$.

15. $(a + b + c)^3$ equals:

(A) $a^3+b^3+c^3+3(a+b)(b+c)(c+a)$ **(B)** $a^3+b^3+c^3+3abc$ **(C)** $a^3+b^3+c^3+3(a+b+c)(ab+bc+ca)$ **(D)** $(a+b)(b+c)(c+a)$

Ans: (A) · Standard identity; option C misses the -3abc correction, so A is the clean exact form.

Chapter 6: Perimeter and Area ★ MOST IMPORTANT

> **Ek line mein chapter**
> Book ka sabse scoring chapter - Heron, circles, sectors, rings aur composite figures. Ek hi baat yaad rakho: figure banao (2 inch ka bhi), values likho, phir formula fire karo. π ka choice (22/7 vs 3.14) pehla decision hai har circle question ka.

6.1 Saare Formulas - Easy Bhasha Mein

Har formula ke saath uska **easy matlab** aur ek **tiny example** hai - formula akela kabhi mat rattro, example ke saath yaad karo. Exam mein pehle 2 second sochho kaunsa formula fire karna hai, phir likho.

Formula (ratt lo)	Easy bhasha mein matlab	Tiny example
Rectangle: P = 2(l + b), A = l × b Square: P = 4a, A = a^2, diagonal = a√2	Basic shapes. Square ka diagonal side ka √2 guna - reverse mein bhi use hota hai.	Square side 7 → diagonal 7√2
Triangle: A = ½ × base × height	Height base ke PERPENDICULAR honi chahiye. Parallelogram ka aadha hota hai same base-height pe.	b = 14, h = 9 → A = 63 cm^2
Heron: s = (a+b+c)/2, A = √(s(s−a)(s−b)(s−c))	Teen sides diye ho toh yehi. Rhythm: s nikalo → 3 differences → multiply (pair smart) → root.	3, 4, 5: s = 6, A = √(6·3·2·1) = 6
Right triangle? A = ½ × leg × leg (Heron SKIP)	Sides 5-12-13, 9-40-41 jaise dikhe ($a^2+b^2=c^2$) toh right angle hai - direct formula, time bachao.	9, 40, 41 → A = 180 cm^2
Equilateral: A = (√3/4)a^2, height = (√3/2)a	Sab sides barabar? Heron ki zaroorat nahi. Reverse bhi: area diya ho toh side = √(4A/√3).	a = 8 → A = 16√3 cm^2
Parallelogram: A = b×h Rhombus: A = ½d_1d_2 Trapezium: A = ½(a+b)h	Rhombus mein DIAGNALS (perpendicular) ka product aadha. Trapezium mein parallel sides ka sum, height, aadha.	d_1=10, d_2=24 → rhombus A = 120
Circle: C = 2πr, A = πr^2	π ka choice: radius 7 ka multiple ho toh 22/7, warna 3.14. Reverse: C diya → r = C/(2π).	r = 14: C = 88 cm, A = 616 cm^2
Arc: ℓ = (θ/360) × 2πr Sector area: (θ/360) × πr^2 = ½ℓr Sector perimeter = ℓ + 2r	Sector = pizza slice: poore circle ka θ/360 hissa. Gem: A = ½ × arc × r. Perimeter mein DONO radii + arc.	r = 21, θ = 60°: ℓ = 22, A = 77
Segment area = sector − triangle	Chord aur arc ke beech ka hissa. Sector se equilateral/60° triangle minus karo (θ=60° pe triangle equilateral hota hai).	θ = 60°, r = 21: segment = 77 − (√3/4)×441

Formula (ratt lo)	Easy bhasha mein matlab	Tiny example
Ring: $A = \pi(R^2 - r^2) = \pi(R+r)(R-r)$ Outside path: $R = r + w$	Donut ka area. Path bahar ho toh outer radius = inner + width; path andar ho toh radii minus karo.	$R = 12$, $r = 8$: $A = \pi \times 20 \times 4 = 320 \times 22/7 \approx 1005.7$
Wheel: distance = revolutions × circumference	Kitne chakkar = kitni distance / circumference. Reverse bhi chalta hai.	$d = 70$ cm wheel, 1 km = 100000 cm → 100000/220 ≈ 455 rev
Brahmagupta (cyclic quadrilateral): $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$	Char sides wala CYCLIC quadrilateral (sab vertices circle par). Same s = semi-perimeter; koi (s−..) multiply hota hai.	$s = 31$, sides 13,14,15,20 type → root before multiply
1 hectare = 10000 m^2	Bade field ka unit. hectare → m^2 mein x10000.	0.5 hectare = 5000 m^2

6.2 Concept Checklist - Ye Sab Aana Hai

- **Heron mein pairing smart karo:** (s−a)(s−b)(s−c) ko aise jodo ki multiplication easy ho - jaise 18×7×11×4 ko (18×4)×(7×11).
- **Wire re-bending questions:** perimeter SAME rehta hai - pehle shape ka perimeter nikaalo, wahi naye shape ka radius/side banta hai.
- **Ratio questions:** sides ko 3x, 4x, 5x likho, perimeter/x diya hua use karke x nikaalo, phir Heron.
- **Composite figures:** rectangle + semicircle, circle in square - area ko TODDO (add ya minus), perimeter ko TRACE karo (sirf outer boundary).
- **Ring ke 2 givens:** (R−r) aur (R+r) alag alag diye ho toh direct (R+r)(R−r) multiply karo - individual radii ki zaroorat nahi.
- **Chord wala ring trick:** chord of OUTER circle tangent to inner → half-chord^2 = $R^2 - r^2$ (Pythagoras) → ring area = $\pi \times (\text{chord}/2)^2$.
- **Goat/cow grazing:** corner pe tied → sector of angle 90°; edge ke beech mein → 180°. Sector area = fraction of circle.
- **Track questions:** perimeter = 2 straights + 1 full circle (2 semicircles); area = rectangle + full circle.
- **Same perimeter different shapes:** circle ka area SABSE BADA hota hai - comparison MCQs ka standard answer.
- **Units discipline:** cm vs m vs km - pehle same unit mein lao, phir formula. Answer ke saath unit likhna compulsory hai.

6.3 Speed Tricks & Trap Alarms

Instant trick	Fire it when...
π choice: radius multiple of 7 → 22/7; otherwise 3.14	First decision of every circle question.
Sector and arc = $\theta/360 \times$ (full circle value). Gem: sector area = ½ × arc × r	Pizza-slice sector questions.
Sides satisfy $a^2 + b^2 = c^2$? Area = ½ × leg × leg - skip Heron	5-12-13, 9-40-41 style triangles.

Instant trick	Fire it when...
Heron rhythm: s → three differences → multiply (pair smartly) → root	Three sides given, height unknown.
Ring: A = π(R+r)(R-r). Path outside: R = r + w. Path inside: subtract squares	Garden path / ring / border questions.
Wheel: distance = revolutions × circumference	"How many revolutions / how far does it travel".
Equilateral: area = $(\sqrt{3}/4)a^2$, height = $(\sqrt{3}/2)a$	All sides equal - no Heron needed.

6.4 Most Important Questions (12)

Ye wo questions hain jo **book ke exercises + exam pattern** se seedhe uthaye gaye hain - inka style, numbers aur twist exam mein aane ke sabse zyada chances hain. Solution chhota hai par poora logic covered hai.

IMP Q1 | Heron direct (T2)

Q1. Find the area of a triangle with sides 5 cm, 12 cm and 13 cm.

Step 1. s = (5 + 12 + 13)/2 = 15 cm.

Step 2. Differences: s - a = 10, s - b = 3, s - c = 2.

Step 3. A = $\sqrt{(15 \times 10 \times 3 \times 2)} = \sqrt{900} = 30$ cm^2.

Step 4. Cross-check: $5^2 + 12^2 = 169 = 13^2$, so it is right-angled; $\frac{1}{2} \times 5 \times 12 = 30$ ✓.

Answer: 30 cm²

Hinglish: *Heron ka 4-step rhythm: s nikaalo, differences nikaalo, multiply, root. 15×10×3×2 = 900, root = 30. Bonus: 5-12-13 Pythagorean triple hai - ½×5×12 = 30 se confirm.*

IMP Q2 | Arc and sector (T7)

Q2. For a circle of radius 7 cm, find (i) the length of an arc subtending 90° at the centre and (ii) the area of the corresponding sector.

Step 1. Fraction of the circle: 90/360 = 1/4.

Step 2. Arc = (1/4) × 2πr = (1/4) × 44 = 11 cm.

Step 3. Sector area = (1/4) × πr^2 = (1/4) × 154 = 38.5 cm^2.

Answer: arc = 11 cm, sector = 38.5 cm²

Hinglish: *90° matlab quarter circle. Arc = 44/4 = 11 cm, area = 154/4 = 38.5 cm². Fraction pehle likho, calculation aadhi ho jati hai.*

IMP Q3 | Ring area (T9)

Q3. Two concentric circles have radii 10 cm and 6 cm. Find the area of the ring between them ($\pi = 3.14$).

Step 1. Ring area = $\pi(R^2 - r^2) = \pi(100 - 36) = 64\pi$.

Step 2. With $\pi = 3.14$: $64 \times 3.14 = 200.96$ cm^2.

Answer: 200.96 cm²

Hinglish: *Ring = bada circle - chhota circle. $\pi(100-36) = 64\pi$. 3.14 se: 200.96 cm². R^2-r^2 ko $(R+r)(R-r)$ likhoge toh mental math ho jata hai.*

IMP Q4 | Heron classic (T2)

Q4. A triangular park has sides 122 m, 120 m and 22 m. Find its area.

Step 1. $s = (122 + 120 + 22)/2 = 132$ m.

Step 2. Differences: 132-122 = 10, 132-120 = 12, 132-22 = 110.

Step 3. Product: $132 \times 10 \times 12 \times 110 = 1{,}742{,}400$.

Step 4. $A = \sqrt{1742400} = 1320$ m^2.

Answer: 1320 m²

Hinglish: *s = 132, differences 10, 12, 110. Product 1742400, root 1320 m². Note the beauty: root = s × 10 = 1320 - aise patterns pehchano.*

IMP Q5 | Brahmagupta (T5)

Q5. A cyclic quadrilateral has sides 13 cm, 10 cm, 13 cm and 10 cm (in order). Find its area using Brahmagupta's formula.

Step 1. Cyclic quadrilateral ⇒ Brahmagupta applies: $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$.

Step 2. $s = (13 + 10 + 13 + 10)/2 = 23$ cm.

Step 3. Differences: 10, 13, 10, 13.

Step 4. Product: $10 \times 13 \times 10 \times 13 = 16900$. $A = \sqrt{16900} = 130$ cm^2.

Answer: 130 cm²

Hinglish: *Cyclic quadrilateral mein Heron ka bhai Brahmagupta: same (s - side) pattern, char baar. s = 23, differences 10,13,10,13. Product 16900, root 130 cm².*

IMP Q6 | Wheel revolutions (T13)

Q6. A wheel of diameter 70 cm makes a journey of 1.1 km. How many complete revolutions does it make ($\pi = 22/7$)?

Step 1. Circumference = $\pi d = (22/7) \times 70 = 220$ cm.

Step 2. Convert the journey: 1.1 km = 1100 m = 110000 cm.

Step 3. Revolutions = 110000/220 = 500.

Answer: 500 revolutions

Hinglish: *Ek chakkar = circumference = 220 cm. Distance 110000 cm. 110000/220 = 500 revolutions. Units same karo pehle - km vs cm sabse common galti.*

IMP Q7 | Sector area and arc (T7)

Q7. A sector of a circle of radius 21 cm has central angle 120°. Find (i) the arc length, (ii) the sector area, (iii) the sector's perimeter.

Step 1. Fraction: 120/360 = 1/3.

Step 2. Full circle: $C = 2\pi(21) = 132$ cm; $A = \pi(441) = 1386$ cm^2.

Step 3. Arc = 132/3 = 44 cm.

Step 4. Sector area = 1386/3 = 462 cm^2.

Step 5. Perimeter = arc + 2 radii = 44 + 42 = 86 cm.

Answer: arc 44 cm; area 462 cm^2; perimeter 86 cm

Hinglish: *120° = one-third. Arc 44, area 462. Perimeter mein DO radii: 44 + 42 = 86 cm. Arc + ek radius likhna classic mistake hai.*

IMP Q8 | Wire re-bending (T12)

Q8. A wire of length 132 cm is first bent into an equilateral triangle and then re-bent into a circle. Which shape encloses more area, and by how much ($\sqrt{3} \approx 1.73$, $\pi = 22/7$)?

Step 1. Equilateral triangle: side = 132/3 = 44 cm.

Step 2. Area = $(\sqrt{3}/4)(44)^2 = (\sqrt{3}/4)(1936) = 484\sqrt{3} \approx 837.3$ cm^2.

Step 3. Circle: circumference 132 $\Rightarrow$ r = 132 × 7/44 = 21 cm.

Step 4. Area = $\pi(441) = 1386$ cm^2.

Step 5. Difference = 1386 - 837.3 = 548.7 cm^2. The circle wins.

Answer: Circle: 1386 vs $484\sqrt{3} \approx 837.3$; the circle encloses about 548.7 cm^2 more.

Hinglish: *Same wire, do shapes. Triangle: $484\sqrt{3} \approx 837$ cm^2. Circle: 1386 cm^2. Circle 549 cm^2 zyada. Ye universal truth hai - fixed perimeter par circle sabse bada area deta hai.*

IMP Q9 | Outside circular path (T10)

Q9. A circular park of radius 17 m is surrounded by a path 4 m wide running outside it. Find the area of the path ($\pi = 22/7$).

Step 1. Outer radius R = 17 + 4 = 21 m.

Step 2. Path area = $\pi(R^2 - r^2) = (22/7)(441 - 289) = (22/7)(152)$.

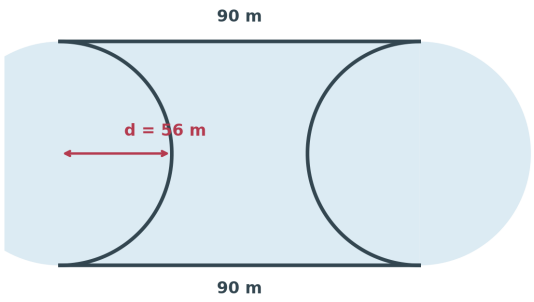
Step 3. = $3344/7 \approx 477.7$ m^2.

Answer: $152\pi \approx 477.7$ m^2

Hinglish: *Bahar wala path: R = 21. $\pi(152) \approx 477.7$ m^2. Shortcut: $\pi w(2r+w) = (22/7)\times4\times38$ - same answer, direct.*

IMP Q10 | Running track (T11)

Q10. A running track is 90 m long on each straight and has two semicircular ends of diameter 56 m. Find (i) the perimeter of the track and (ii) the area enclosed ($\pi = 22/7$).



Step 1. Perimeter = two straights + two semicircular arcs = 90 + 90 + one full circle of diameter 56.

Step 2. Circle circumference = $\pi d = (22/7)(56) = 176$ m.

Step 3. Perimeter = 180 + 176 = 356 m.

Step 4. Area = rectangle 90×56 + full circle $\pi(28^2) = 5040 + (22/7)(784) = 5040 + 2464 = 7504$ m^2.

Answer: (i) 356 m (ii) 7504 m^2

Hinglish: *Do semicircles = ek full circle. Perimeter 180 + 176 = 356 m. Area 5040 + 2464 = 7504 m^2. Track problems mein halves ko jod kar full banao.*

IMP Q11 | Costs: fence and plough (T13)

Q11. A circular garden has radius 35 m. Fencing costs ₹20 per metre and ploughing costs ₹0.50 per m^2. Find the total cost of fencing the boundary once and ploughing the whole garden ($\pi = 22/7$).

Step 1. Circumference = $2\pi r = 2 \times (22/7) \times 35 = 220$ m.

Step 2. Fencing cost = 220×20 = ₹4400.

Step 3. Area = $\pi r^2 = (22/7)(1225) = 3850$ m^2.

Step 4. Ploughing cost = 3850 × 0.50 = ₹1925.

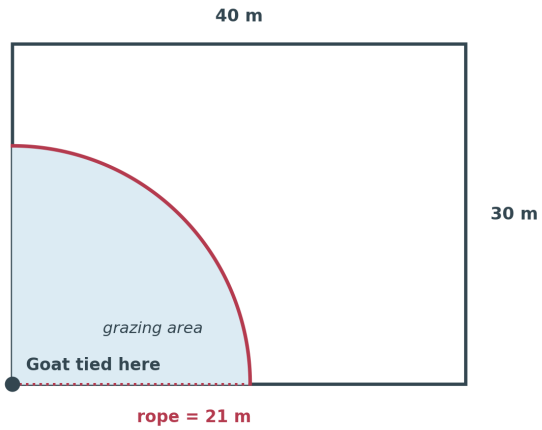
Step 5. Total = 4400 + 1925 = ₹6325.

Answer: ₹6325

Hinglish: *Fencing = boundary × rate/m: 220×20 = 4400. Ploughing = area × rate/m²: 3850×0.5 = 1925. Total ₹6325. Rate ke saath unit dekho - /m ya /m².*

IMP Q12 | Grazing goat (T13/T14)

Q12. A goat is tied to one corner of a rectangular field 40 m by 30 m with a rope 21 m long. Find the area it can graze, assuming it cannot leave the field (π = 22/7).



Step 1. At a corner of a rectangle, only 90° of the circle's sweep lies inside the field: a quarter circle.

Step 2. Grazing area = $(1/4)\pi r^2 = (1/4) \times (22/7) \times 441 = 346.5$ m^2.

Answer: 346.5 m²

Hinglish: *Corner par sirf 90° ka sector available - quarter circle. ¼ × π × 441 = 346.5 m². Rope = radius, corner = center.*

6.5 Rapid-Fire MCQs (15)

Rapid fire: option padho, dimaag mein answer banao, phir Answer line se check karo. Galat hua toh Answer line ka logic padho - wahi 1 line exam mein marks bachayegi.

1. A rectangle is 15 cm by 8 cm. Its perimeter and area are:

(A) 46 cm, 120 cm^2 **(B)** 23 cm, 120 cm^2 **(C)** 46 cm, 100 cm^2 **(D)** 30 cm, 120 cm^2

Ans: (A) · P = 2(15+8) = 46 cm; A = 15×8 = 120 cm^2.

2. The circumference of a circle of radius 7 cm is:

(A) 44 cm **(B)** 154 cm **(C)** 22 cm **(D)** 88 cm

Ans: (A) · $2\pi r = 2\times(22/7)\times7 = 44$ cm.

3. The area of a circle of diameter 14 cm is:

(A) 154 cm^2 **(B)** 616 cm^2 **(C)** 44 cm^2 **(D)** 88 cm^2

Ans: (A) · $r = 7$: $\pi r^2 = 154$ cm^2.

4. The area of an equilateral triangle of side 8 cm is:

(A) $16\sqrt{3}$ cm^2 **(B)** $32\sqrt{3}$ cm^2 **(C)** $8\sqrt{3}$ cm^2 **(D)** $64\sqrt{3}$ cm^2

Ans: (A) · $(\sqrt{3}/4)(64) = 16\sqrt{3}$.

5. The area of a triangle with sides 3, 4, 5 cm is:

(A) 6 cm^2 **(B)** 12 cm^2 **(C)** 10 cm^2 **(D)** 7.5 cm^2

Ans: (A) · Right triangle ($3^2+4^2 = 5^2$): $\frac{1}{2}\times3\times4 = 6$ cm^2.

6. The area of a sector of angle 90° in a circle of radius 14 cm is:

(A) 154 cm^2 **(B)** 77 cm^2 **(C)** 38.5 cm^2 **(D)** 616 cm^2

Ans: (A) · $(1/4)\pi(196) = 154$ cm^2.

7. Two concentric circles have radii 7 cm and 14 cm. The ring area is:

(A) 462 cm^2 **(B)** 154 cm^2 **(C)** 616 cm^2 **(D)** 231 cm^2

Ans: (A) · $\pi(196 - 49) = \pi\cdot147 = 462$ cm^2.

8. A trapezium has parallel sides 10 cm and 14 cm, height 6 cm. Its area is:

(A) 72 cm^2 **(B)** 84 cm^2 **(C)** 60 cm^2 **(D)** 48 cm^2

Ans: (A) · $\frac{1}{2}(10+14)(6) = 72$ cm^2.

9. The area of a triangle with sides 13, 14, 15 cm is:

(A) 84 cm^2 **(B)** 91 cm^2 **(C)** 78 cm^2 **(D)** 105 cm^2

Ans: (A) · $s = 21$; $21\cdot6\cdot7\cdot8 = 7056$; $\sqrt{7056} = 84$.

10. A wheel of diameter 70 cm travels 2.2 km. The number of revolutions is:

(A) 1000 **(B)** 500 **(C)** 2000 **(D)** 1100

Ans: (A) · $C = 220$ cm; $220000/220 = 1000$.

11. A cyclic quadrilateral has sides 8, 15, 8, 15 cm. Its area is:

(A) 120 cm^2 **(B)** 115 cm^2 **(C)** 46 cm^2 **(D)** 240 cm^2

Ans: (A) · $s = 23$; $(23-8)^2(23-15)^2 = 15^2 \cdot 8^2$; area = 120.

12. The radii of two circles are in the ratio 2 : 5. The ratio of their areas is:

(A) 4 : 25 **(B)** 2 : 5 **(C)** 4 : 10 **(D)** 8 : 125

Ans: (A) · Areas carry the square of the ratio: $2^2 : 5^2 = 4 : 25$.

13. An arc subtends 120° at the centre of a circle of radius 21 cm. Its length is:

(A) 44 cm **(B)** 22 cm **(C)** 66 cm **(D)** 88 cm

Ans: (A) · $(1/3)(132) = 44$ cm.

14. The area swept by a 21 cm long minute hand in 10 minutes is ($\pi = 22/7$):

(A) 231 cm^2 **(B)** 1386 cm^2 **(C)** 77 cm^2 **(D)** 115.5 cm^2

Ans: (A) · 10 min → 60° → $(1/6)\pi(441) = 231$ cm^2.

15. A square and a circle have EQUAL areas of 154 cm^2 ($\pi = 22/7$). Which is true about their perimeters?

(A) The square's perimeter is larger **(B)** The circle's perimeter is larger **(C)** They are equal **(D)** Cannot be determined

Ans: (A) · Circle r = 7, C = 44. Square side $\sqrt{154} \approx 12.41$, $P \approx 49.64 > 44$.

Chapter 7: Probability

> **Ek line mein chapter**
> Chance ka math - coin, dice, cards, bags. Sab kuch ek fraction hai: favourable/total. Bas total outcomes ginna aana chahiye (2 dice = 36, 3 coins = 8, deck ke 5 magic numbers). "At least one" dikhe toh complement laga do - aadhe questions wahi se solve hote hain.

7.1 Saare Formulas - Easy Bhasha Mein

Har formula ke saath uska **easy matlab** aur ek **tiny example** hai - formula akela kabhi mat rattro, example ke saath yaad karo. Exam mein pehle 2 second sochho kaunsa formula fire karna hai, phir likho.

Formula (ratt lo)	Easy bhasha mein matlab	Tiny example
P(E) = favourable outcomes / total outcomes	Probability = jitne result hum chahte hain / sab possible results. Hamesha 0 se 1 ke beech.	Die pe even: 3/6 = 1/2
$0 \le P(E) \le 1$; sure event P = 1; impossible P = 0	Probability 1.4 ya negative kabhi nahi ho sakti. MCQ mein "which CANNOT be a probability" - 1.4 type options.	P(sun rises) = 1; P(7 on a die) = 0
P(not E) = 1 − P(E)	Complement: "nahi hone ka chance" = 1 minus hone ka chance. "At least one" ke liye GOLD: P(at least one) = 1 − P(none).	P(E) = 0.65 → P(not E) = 0.35
n coins → 2^n outcomes 1 exact sequence (HHT) = $1/2^n$	2 coins → 4, 3 coins → 8 outcomes. "Exactly HHT" jaise kisi EK order ka chance 1/8 hota hai.	3 coins exactly 2 heads: 3/8 (HHt, HtH, tHH)
1 die → 6 outcomes; 2 dice → 36 (ordered pairs)	Do dice ko (first, second) jodi mein gino - 6 × 6 = 36. Sum 7 raja hai: 6 ways (1-6, 2-5, 3-4, 4-3, 5-2, 6-1).	P(sum = 7) = 6/36 = 1/6
Cards ke 5 magic numbers: 4 aces, 12 face, 26 red, 13 per suit, 9 numbered per suit	Deck ke 90% answers inhi 5 numbers se aate hain. Ratta maar lo - card question mein sochna hi nahi padega.	P(ace) = 4/52 = 1/13; P(face) = 12/52 = 3/13
Without replacement: numerator AUR denominator dono shrink	Ball/card wapas nahi mila toh dusri draw mein total ek kam, favourable bhi ho sakta hai ek kam.	2 reds from 5R3B: (5/8)×(4/7) = 20/56 = 5/14
Empirical P = times it happened / total trials	Experiment/data se probability. 400 mein 96 walk → P = 96/400 = 0.24.	500 bulbs, 15 defective → P(good) = 485/500 = 0.97
Geometric P = favourable area / total area	Dartboard, spinner, segment - chance = area ka fraction. Angles bhi area hi hain (spinner).	Inner circle r=5 in 20×20 board: 25π/400
Mutually exclusive: P(A ya B) = P(A) + P(B)	Dono events ek saath kabhi nahi ho sakte (red ya king jaise overlap ho toh common minus karo).	P(red or king) = 26/52 + 4/52 − 2/52 = 28/52

7.2 Concept Checklist - Ye Sab Aana Hai

- **Sample space pehle likho** - S = {1,2,3,4,5,6} ya {HH, HT, TH, TT}. Bina S ke answer likhna risky hai.
- **Coins mein HT ≠ TH galti:** unordered sochoge toh 3 outcomes aayenge - GALAT. Hamesha ordered gino (4, 8).
- **Two-dice grid banao:** 6×6 table mein sum/product bharna 30 second leta hai aur kabhi nahi galat hota.
- **"At least one six" in 2 dice** = 1 − P(no six) = 1 − $(5/6)^2$ = 11/36. Direct ginte waqt 6 pairs bhool jaate hain!
- **Difference events:** |a−b| = 1 ke 10 ways hote hain ((1,2)...(6,5) dono order) - ordered pair hi rakhna.
- **Empirical table:** saare P ka total EXACTLY 1 hona chahiye - missing probability isi se nikalti hai.
- **Same colour / different colour draws:** RR + BB + RB jaise cases ko add karo, ya ek hi colour ke pairs gino.
- **Game fairness:** dono ke total chances barabar = fair. P(A) + P(B) + P(draw) = 1 yaad rakho.
- **Expected count:** 300 rolls pe six ≈ 300 × 1/6 = 50 times. P × trials = expectation.

7.3 Speed Tricks & Trap Alarms

Instant trick	Fire it when...
"At least one" → 1 - P(none). Works for coins, dice, bags	Any "at least one" wording.
Without replacement: numerator AND denominator both shrink	Two or three draws, balls not returned.
Deck counts: 4 aces, 12 face cards, 26 red, 13 per suit, 9 numbered per suit	Every card question - 5 numbers, 90% of answers.
Two dice = 36 ordered pairs; sum 7 is the king (6 ways). Draw the 6×6 grid	Sum / product / difference on two dice.
n coins → 2^n outcomes; one exact sequence (like HHT) = $1/2^n$	Three-coins questions.
Geometric P = favourable area / total area	Darts, spinners, random points on segments.
P(E) + P(not E) = 1	"Find P(not ...)" or when P(E) is given directly.

7.4 Most Important Questions (10)

Ye wo questions hain jo **book ke exercises + exam pattern** se seedhe uthaye gaye hain - inka style, numbers aur twist exam mein aane ke sabse zyada chances hain. Solution chhota hai par poora logic covered hai.

IMP Q1 | Two coins (T1/T8)

Q1. Two coins are tossed together. Find P(i) exactly one head, (ii) at least one head, (iii) no head.

Step 1. S = {HH, HT, TH, TT}: n(S) = 4.

Step 2. (i) Exactly one head: {HT, TH}: P = 2/4 = 1/2.

Step 3. (ii) At least one head: {HH, HT, TH}: P = 3/4.

Step 4. (iii) No head: {TT}: P = 1/4. Note (ii) and (iii) are complements: 3/4 + 1/4 = 1 ✓.

Answer: (i) 1/2 (ii) 3/4 (iii) 1/4

Hinglish: *Chaar outcomes. Exactly one head = HT, TH = 1/2. At least one = 3/4 (complement of TT). At least wale questions mein 'none' nikaalo, 1 minus karo.*

IMP Q2 | Single draw from a bag (T6)

Q2. A bag contains 4 red, 5 blue and 3 green balls. One ball is drawn at random. Find P(i) red, (ii) not green, (iii) red or blue.

Step 1. Total = 4 + 5 + 3 = 12.

Step 2. (i) P(red) = 4/12 = 1/3.

Step 3. (ii) P(not green) = 1 - 3/12 = 9/12 = 3/4.

Step 4. (iii) P(red or blue) = (4+5)/12 = 9/12 = 3/4 (which matches (ii), since 'not green' = 'red or blue').

Answer: (i) 1/3 (ii) 3/4 (iii) 3/4

Hinglish: *Total 12. Red 1/3. Not green 3/4. Red-or-blue bhi 3/4 - same cheez hai dono! Consistency check hamesha karo.*

IMP Q3 | Card basics (T4)

Q3. One card is drawn from a well-shuffled deck of 52. Find P(i) an ace, (ii) a red card, (iii) a face card, (iv) the queen of hearts.

Step 1. (i) Aces: 4 → P = 4/52 = 1/13.

Step 2. (ii) Red cards: 26 → P = 26/52 = 1/2.

Step 3. (iii) Face cards (J, Q, K): 3 × 4 = 12 → P = 12/52 = 3/13.

Step 4. (iv) One specific card: P = 1/52.

Answer: (i) 1/13 (ii) 1/2 (iii) 3/13 (iv) 1/52

Hinglish: *Yaad rakho: aces 4, reds 26, face cards 12, hearts 13. 1/13, 1/2, 3/13, 1/52. Ye counts ratta - deck questions 90% inhi par.*

IMP Q4 | Two-dice sum intro (T3)

Q4. Two dice are rolled together. Find P(sum = 7) and P(sum = 2).

Step 1. n(S) = 36 ordered pairs.

Step 2. Sum 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1): 6 pairs → P = 6/36 = 1/6.

Step 3. Sum 2: (1,1) only → P = 1/36.

Answer: 1/6 and 1/36

Hinglish: *36 outcomes. Sum 7 six tarike - sabse likely! 1/6. Sum 2 sirf (1,1) - 1/36. Grid banao rough mein, pairs ginn lo.*

IMP Q5 | Two-dice conditions (T3)

Q5. Two dice are rolled. Find P(i) the sum is a prime number, (ii) the product is odd, (iii) both dice show the same number.

Step 1. (i) Prime sums: 2 (1 pair), 3 (2), 5 (4), 7 (6), 11 (2): total 15 pairs → P = 15/36 = 5/12.

Step 2. (ii) Product odd ⇔ both dice odd: 3 × 3 = 9 pairs → P = 9/36 = 1/4.

Step 3. (iii) Doubles: (1,1), ..., (6,6): 6 pairs → P = 6/36 = 1/6.

Answer: (i) 5/12 (ii) 1/4 (iii) 1/6

Hinglish: *Prime sums ka count: 1+2+4+6+2 = 15 → 5/12. Product odd = dono odd = 1/4. Doubles 1/6. 36-grid se teeno direct padho.*

IMP Q6 | Three coins (T8)

Q6. Three coins are tossed together. Find P(i) exactly two heads, (ii) at least two heads, (iii) all three same.

Step 1. S has $2^3 = 8$ outcomes.

Step 2. (i) Exactly two heads: HHT, HTH, THH → 3/8.

Step 3. (ii) At least two heads: those 3 plus HHH → 4/8 = 1/2.

Step 4. (iii) All same: HHH, TTT → 2/8 = 1/4.

Answer: (i) 3/8 (ii) 1/2 (iii) 1/4

Hinglish: *8 outcomes. Exactly 2: teen paths = 3/8. At least 2: +HHH = 1/2. All same 1/4. Tree diagram banao - 8 paths dikh jayengi.*

IMP Q7 | At least one (T5)

Q7. A coin is tossed 4 times (or four coins together). Find P(at least one tail).

Step 1. Total outcomes: $2^4 = 16$.

Step 2. Complement: no tail means all heads: 1 outcome.

Step 3. P(at least one tail) = 1 - 1/16 = 15/16.

Answer: 15/16

Hinglish: *At least one tail = 1 - (no tail). No tail = HHHH = 1/16. Answer 15/16. Complement rule ka best use.*

IMP Q8 | Cards: two aces (T7)

Q8. Two cards are drawn from a deck without replacement. Find P(both are aces).

Step 1. First ace: 4/52 = 1/13. Second ace: 3/51 = 1/17.

Step 2. P = (1/13)(1/17) = 1/221.

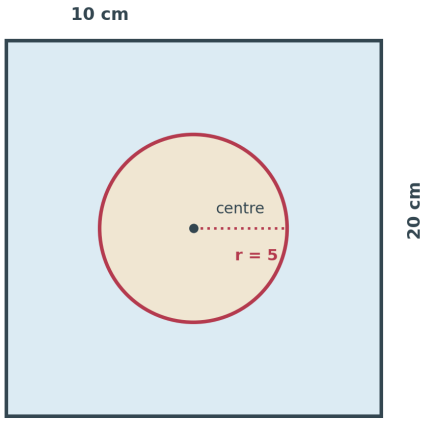
Step 3. Combination check: C(4,2)/C(52,2) = 6/1326 = 1/221 ✓.

Answer: 1/221

Hinglish: *1/13 × 1/17 = 1/221. Doosra ace 51 cards mein se 3 - shrinking deck.*

IMP Q9 | Geometric: dartboard (T10)

Q9. A dart hits a square board of side 20 cm at a random point. A circle of diameter 10 cm is painted at the centre. Find P(the dart lands inside the circle).



Step 1. Board area = 400 cm^2. Circle radius = 5 cm; area = 25π cm^2.

Step 2. P = 25π/400 = π/16.

Step 3. With π = 22/7: 22/112 = 11/56 ≈ 0.196.

Answer: π/16 ≈ 0.196

Hinglish: *Circle/board = 25π/400 = π/16. Dono same measure (area) - yahi rule hai. π symbolic rakho jab tak ho sake.*

IMP Q10 | At least one with bag draws (T5/T7)

Q10. A bag has 4 red and 4 blue balls. Three balls are drawn without replacement. Find P(at least one red).

Step 1. Complement: no red = all three blue.

Step 2. P(all blue) = (4/7)(3/6)(2/5) = 24/210 = 4/35.

Step 3. P(at least one red) = 1 - 4/35 = 31/35.

Answer: 31/35

Hinglish: *At least one red = 1 - (sab blue). Sab blue: 4/7×3/6×2/5 = 4/35. Answer 31/35. Complement hi sahi raasta hai.*

7.5 Rapid-Fire MCQs (13)

Rapid fire: option padho, dimaag mein answer banao, phir Answer line se check karo. Galat hua toh Answer line ka logic padho - wahi 1 line exam mein marks bachayegi.

1. A die is rolled once. P(a number less than 3) is:

(A) 1/3 **(B)** 1/2 **(C)** 2/3 **(D)** 1/6

Ans: (A) · {1, 2}: 2/6 = 1/3.

2. Two coins are tossed. P(exactly two heads) is:

(A) 1/4 **(B)** 1/2 **(C)** 3/4 **(D)** 1/3

Ans: (A) · Only HH among 4 outcomes: 1/4.

3. A die is rolled. 'Getting a number less than 10' has probability:

(A) 1 **(B)** 5/6 **(C)** 0 **(D)** 9/10

Ans: (A) · Every outcome satisfies it: 6/6 = 1 - a sure event.

4. One card is drawn from 52. P(an ace) is:

(A) 1/13 **(B)** 1/4 **(C)** 1/52 **(D)** 4/13

Ans: (A) · 4 aces: 4/52 = 1/13.

5. P(a face card) from a deck of 52 is:

(A) 3/13 **(B)** 1/13 **(C)** 4/13 **(D)** 12/13

Ans: (A) · 12 face cards: 12/52 = 3/13.

6. Two dice: P(sum = 7) is:

(A) 1/6 **(B)** 1/36 **(C)** 7/36 **(D)** 5/36

Ans: (A) · Six pairs sum to 7: 6/36 = 1/6.

7. Which CANNOT be a probability?

(A) 1.4 **(B)** 0 **(C)** 1 **(D)** 0.999

Ans: (A) · Probabilities lie in [0, 1]; 1.4 exceeds 1.

8. P(E) = 0.65. P(not E) equals:

(A) 0.35 **(B)** 0.65 **(C)** 1.65 **(D)** 0.5

Ans: (A) · 1 - 0.65 = 0.35.

9. Three coins: P(exactly two heads) is:

(A) 3/8 **(B)** 1/2 **(C)** 1/8 **(D)** 5/8

Ans: (A) · HHT, HTH, THH: 3 of 8.

10. Two dice: P(at least one 6) is:

(A) 11/36 **(B)** 1/6 **(C)** 1/36 **(D)** 25/36

Ans: (A) · Complement: no six = 25/36; so 11/36.

11. A spinner has equal quarters marked A, B, C, D. P(not A) is:

(A) 3/4 **(B)** 1/4 **(C)** 1/2 **(D)** 1/3

Ans: (A) · 1 - 1/4 = 3/4.

12. Three coins: P(at least one head) is:

(A) 7/8 **(B)** 3/8 **(C)** 1/2 **(D)** 15/16

Ans: (A) · 1 - P(all tails) = 1 - 1/8 = 7/8.

13. A die is rolled 3 times. P(at least one 6) is:

(A) 91/216 **(B)** 1/2 **(C)** 3/6 **(D)** 125/216

Ans: (A) · $1 - (5/6)^3 = 1 - 125/216 = 91/216$.

Chapter 8: Sequences and Progressions

Ek line mein chapter

Numbers ki lines: AP mein fixed ADD, GP mein fixed MULTIPLY, Virahanka mein pichhle do ka SUM. Formula daalo, term nikaalo, "which term" solve karo. Sierpinski aur doubling wale GP questions kal exam ke favourite hain.

8.1 Saare Formulas - Easy Bhasha Mein

Har formula ke saath uska **easy matlab** aur ek **tiny example** hai - formula akela kabhi mat rattro, example ke saath yaad karo. Exam mein pehle 2 second sochho kaunsa formula fire karna hai, phir likho.

Formula (ratt lo)	Easy bhasha mein matlab	Tiny example
AP: $t_n = a + (n-1)d$, jahan $d = t_2 - t_1$	Har baar utna hi ADD ho raha ho → AP. a = pehla term, d = common difference (minus bhi ho sakta hai!).	3, 8, 13, ...: a = 3, d = 5; $t_{10} = 3 + 45 = 48$
"k is a term?" → solve $a + (n-1)d = k$	n NIKAL ke dekho: n natural number aaya = YES (position n); fraction aaya = NO, kabhi nahi aayega.	$t_n = 3n-7$, k = 557: n = 188 → YES
Sum: $1 + 2 + ... + n = n(n+1)/2$	Gauss ka pairing trick: pehla + aakhri, dobara pehla+1 + aakhri−1 ... har pair same sum, pairs = n/2.	$S_{100} = 100 \times 101/2 = 5050$
Range sum $k + ... + m = S_m - S_{k-1}$ ya ((first + last)/2) × count	Beech ka sum? Dono method same answer. Count = (last − first)/step + 1 bhoolna mat.	25 + ... + 58: (25+58)/2 × 34 = 1411
GP: $t_n = a \cdot r^{n-1}$, jahan $r = t_2/t_1$	Har baar fixed GUNA (multiply/divide) → GP. r negative ho toh terms alternate karti hain.	3, 12, 48: r = 4; $t_5 = 3 \times 4^4 = 768$
Virahanka-Fibonacci: $V_n = V_{n-1} + V_{n-2}$	Pichhle DO terms ka sum agla term. 1, 2, 3, 5, 8, 13, 21, 34, ... Poet Virahanka (7th century) ne diya tha.	13 ke baad: 8+13 = 21, phir 34, 55, 89
Explicit vs Recursive rule	Explicit: kisi bhi n ke liye direct formula ($t_n = 3n + 1$) - jump far. Recursive: agla term pichhle se ($t_{n+1} = t_n + 3$) - ek-ek karke.	Dono same sequence de sakte hain!
Sierpinski: triangles = 3^n, black area = $(3/4)^n$ Carpet: pieces = 8^n, area = $(8/9)^n$	Stage 0 se shuru hota hai (original shape), isliye exponent = stage number. Count GP (r=3 ya 8), area shrinking GP (r=3/4, 8/9).	Stage 4: $3^4 = 81$ triangles, area $(3/4)^4 = 81/256$
Bouncing/folding GP: exponent = number of bounces/stages	Ball 24 ft se giri, har bounce 3/4: heights 18, 13.5, 13.5×3/4, ... a = 18, r = 3/4. Bacteria double: 30×2^n.	20 folds: 0.1 mm × $2^{20} \approx 105$ km? Nahi - 0.1×2^{20} mm = 104.8 m
AP graph = straight line (slope d); GP graph = curve	Points $(1, t_1)$, $(2, t_2)$, ... plot karo: equal steps → line; guna hota gaya → curve. Graph se AP/GP pehchano.	(1,2),(2,5),(3,8) → line $y = 3x-1$

8.2 Concept Checklist - Ye Sab Aana Hai

- **Pehla sawaal har sequence ka:** difference constant hai? → AP. Ratio constant? → GP. Dono nahi → neither (ya quadratic/Virahanka).
- **d nikalte waqt order:** d = baad wala − pehle wala. 11, 7, 3: d = −4 (decreasing AP).
- **Two conditions = two equations:** "$t_3 = 12$, $t_{50} = 106$" → subtract karo, d milta hai, phir a. Ye book ka starred question hai.
- **$t_{11} = 38$, $t_{16} = 73 \rightarrow 5d = 35 \rightarrow d = 7$, $a = -32$; $t_{31} = a + 30d = 178$.** NCERT starred - pakka aata hai.
- **Multiples counting:** "how many 2-digit multiples of 3" → AP 12, 15, ..., 99; n = (99−12)/3 + 1 = 30; sum = n/2 × (first+last).
- **Three numbers in AP?** Unhe a−d, a, a+d likho - "sum diya" se a turant milta hai. GP mein a/r, a, ar.
- **Sum of first n odd numbers = n^2**; first n even = n(n+1); squares 1,4,9 mein differences = odd numbers. Ye identities proof questions mein aati hain.
- **Recursive rule iterate karo dheere se** - har term ko pichhle se banao, jump mat maaro. Hidden pattern dhundhna ho toh 4-5 terms likho.
- **100 as consecutive sums (starred):** 18+...+22 aur 9+...+16 - do hi tarike. Method: n terms → sum = n·(first+last)/2 = 100 solve karo.
- **Exponents count karo:** doubling kitni baar hua? 2^{20} mein 20 doublings. Chessboard 64th square = 2^{63} grains.

8.3 Speed Tricks & Trap Alarms

Instant trick	Fire it when...
Constant +/- between terms? It is an AP: $d = t_2 - t_1$, $t_n = a + (n - 1)d$. 10th term = a + 9d	'Each time 5 more', fixed fee + per-km charges, yearly increments, row/seat counts
Constant x or divide between terms? It is a GP: $r = t_2/t_1$, $t_n = ar^{n-1}$. t_{10} of 3, 6, 12 = 3 x 2^9 = 1536	'Doubles', 'halves', '3/4 of previous height', '10% growth', 'splits into 3'
'Is k a term?' - solve $t_n = k$. n a natural number = YES at position n; n fractional = NO	'Is 100 a term?', 'which term is 607?', 'prove 461 is not a term'
Two term-facts given? Two equations: $a + (n - 1)d$ = value (AP) or ar^k = value (GP). Subtract / divide them	'3rd term is 12 and 50th is 106', '$t_{11} = 38$, $t_{16} = 73$', '$t_2 = 9$, $t_5 = 243$'
Sums: $1 + ... + n = n(n + 1)/2$. Range $k + ... + m = S_m - S_{k-1}$, or (first + last)/2 x count	'Total marbles', 'sum of all multiples of 3 up to 99', '25 + 26 + ... + 58'
Real-life GP: the exponent counts the doublings / stages / bounces. Sierpinski starts at Stage 0, so count = 3^n there	'Bacteria double every hour', 'bounces to 3/4', 'Stage 6 of the fractal', 'folded 20 times'
Graph fingerprint: AP points lie on a straight line of slope d; GP points bend on a steepening curve ($r > 1$) or melt to 0 ($0 < r < 1$)	'Do the points lie on a line?', 'what shape does the plot make?', 'sketch the growth'

8.4 Most Important Questions (11)

Ye wo questions hain jo **book ke exercises + exam pattern** se seedhe uthaye gaye hain - inka style, numbers aur twist exam mein aane ke sabse zyada chances hain. Solution chhota hai par poora logic covered hai.

IMP Q1 | Membership test (T4)

Q1. For the sequence $t_n = 3n - 7$, check whether 557 is a term. Also check whether 460 is a term. Justify both answers.

Step 1. Test 557: solve $3n - 7 = 557$, so $3n = 564$ and $n = 564/3 = 188$.

Step 2. $n = 188$ is a natural number, so 557 IS a term - exactly the 188th.

Step 3. Test 460: solve $3n - 7 = 460$, so $3n = 467$ and $n = 467/3 = 155.666...$

Step 4. n is not a whole number, so 460 is NOT a term of this sequence.

Answer: 557 is a term (the 188th); 460 is not a term (n = 467/3 is not natural).

Hinglish: *Membership test ka funda: equation solve karo, n whole number aaya toh haan, fraction aaya toh na. 557 ke liye n = 188 perfect. 460 ke liye n = 155.67 - aisa position exist hi nahi karta. Bas yahi logic exam mein 2 mark dilaata hai.*

IMP Q2 | AP terms via a + (n-1)d (T8)

Q2. For the AP 3, 8, 13, 18, ...: (i) state a and d. (ii) Find the 10th and 26th terms. (iii) Write the explicit rule t_n.

Step 1. (i) $a = 3$; $d = 8 - 3 = 5$ (verify: $13 - 8 = 5$ ✓).

Step 2. (ii) $t_{10} = 3 + (10 - 1) \times 5 = 3 + 45 = 48$.

Step 3. $t_{26} = 3 + (26 - 1) \times 5 = 3 + 125 = 128$.

Step 4. (iii) $t_n = 3 + (n - 1) \times 5 = 5n - 2$.

Step 5. Cross-check with the previous question: $5n - 2$ with $t_{10} = 48$ and $t_{15} = 72$ matches this AP - same sequence, two descriptions.

Answer: (i) a = 3, d = 5 (ii) $t_{10} = 48$, $t_{26} = 128$ (iii) $t_n = 5n - 2$

Hinglish: *a pehla term, d step size. 10th term = 3 + 9 baar step 5 = 48. Aur $t_n = 5n - 2$ - dekho, coefficient 5 hi d hai aur constant a - d = 3 - 5 = -2. Dono form ek hi kahani.*

IMP Q3 | Gauss pairing and S20 (T10)

Q3. Use the pairing method to find $1 + 2 + 3 + ... + 10$. Then use the formula to find S_{20}.

Step 1. Write $S = 1 + 2 + ... + 10$ and again backwards $S = 10 + 9 + ... + 1$.

Step 2. Add column-wise: every column gives 11, and there are 10 columns, so $2S = 110$.

Step 3. Hence $S = 55$.

Step 4. Formula route: $S_n = n(n + 1)/2$, so $S_{20} = 20 \times 21/2 = 210$.

Answer: 1 + ... + 10 = 55; S_{20} = 210.

Hinglish: *Aage se likho, peechhe se likho, jodo - har column ka total 11, das columns, toh sum aadha = 55. Gauss ko school mein isi trick ne legend bana diya tha. S_{20} ke liye seedha formula: 20 × 21/2 = 210.*

IMP Q4 | Which term is 607 (T4/T8)

Q4. Which term of the sequence t_n = 5n - 3 is 607?

Step 1. Set 5n - 3 = 607.

Step 2. 5n = 610.

Step 3. n = 122.

Step 4. n is a natural number, so 607 is the 122nd term of the sequence.

Answer: The 122nd term.

Hinglish: *Seedha equation: 5n - 3 = 607, n = 122. Whole number aaya toh answer pakka. Fraction aata toh answer hota 'not a term' - yahan toh clean 122 nikla.*

IMP Q5 | AP from two conditions (T9)

Q5. An AP consists of 50 terms. Its 3rd term is 12 and its last (50th) term is 106. Find the 29th term.

Step 1. Let the AP have first term a and common difference d.

Step 2. 3rd term: a + 2d = 12.

Step 3. 50th term: a + 49d = 106.

Step 4. Subtract: 47d = 94, so d = 2. Then a = 12 - 2d = 12 - 4 = 8.

Step 5. t_{29} = 8 + 28 × 2 = 64.

Answer: d = 2, a = 8; t_{29} = 64.

Hinglish: *Do equations, do unknowns - classic combo. 47d = 94 se d = 2, wapas daal ke a = 8. 29th term = 64. Ye pattern itna common hai ki exam mein ye question 'free marks' hai agar method yaad ho.*

IMP Q6 | Two-digit multiples of 3: count and sum (T8/T10)

Q6. How many two-digit numbers are divisible by 3? What is the sum of all these two-digit numbers?

Step 1. Two-digit multiples of 3: smallest 12, largest 99. They form the AP 12, 15, 18, ..., 99 with d = 3.

Step 2. Count: 12 + (n - 1) × 3 = 99 gives (n - 1) × 3 = 87, n - 1 = 29, n = 30. So there are 30 such numbers.

Step 3. Sum by the average idea: sum = (first + last)/2 × count = (12 + 99)/2 × 30.

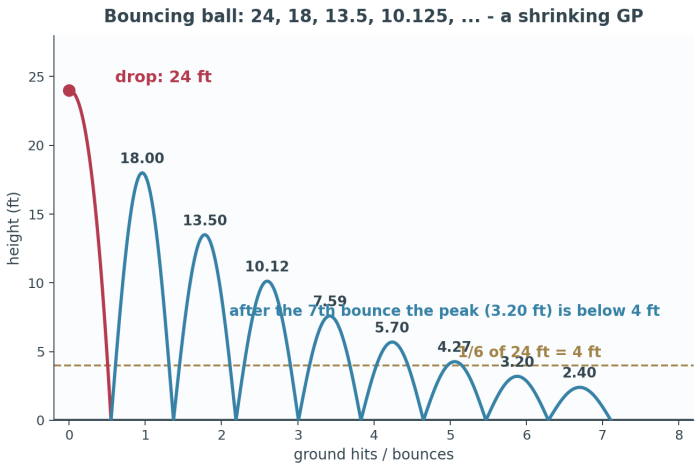
Step 4. = (111/2) × 30 = 111 × 15 = 1665.

Answer: 30 numbers; their sum is 1665.

Hinglish: *12 se 99 tak 3 ke multiples = AP. Count: (99-12)/3 + 1 = 30. Sum ka Āryabhaṭa trick: average × count = 55.5 × 30 = 1665. Pehla-akhirka jodo, aadha karo, count se guna karo - formula rattne ki zaroorat nahi.*

IMP Q7 | Bouncing ball heights (T13)

Q7. A ball is dropped from a height of 24 feet. After hitting the ground it bounces up to 3/4 of its previous height each time. (i) Write the heights reached after each of the first five bounces. (ii) The heights form a GP - state a and r. (iii) Show that after the 7th bounce the ball stays below 1/6 of its original dropping height.



Step 1. (i) Bounce 1: 24 × 3/4 = 18 ft. Bounce 2: 18 × 3/4 = 13.5 ft. Bounce 3: 10.125 ft. Bounce 4: 7.59375 ft. Bounce 5: 5.6953125 ft.

Step 2. (ii) The bounce heights 18, 13.5, 10.125, ... form a GP with first term a = 18 and ratio r = 3/4 (each height is 3/4 of the one before).

Step 3. (iii) 1/6 of 24 ft = 4 ft. Continuing: bounce 6 = 5.6953125 × 3/4 = 4.271484375 ft (still above 4), bounce 7 = 4.271484375 × 3/4 = 3.20361328125 ft.

Step 4. 3.2036... ft < 4 ft, so from the 7th bounce onward the ball never rises above 1/6 of the original height.

Step 5. Formula route: bounce-n height = $18 \times (3/4)^{n-1}$; set it below 4 and count the bounces.

Answer: (i) 18, 13.5, 10.125, 7.59375, 5.6953 ft (ii) a = 18, r = 3/4 (iii) 7th-bounce height 3.2036 ft is below 4 ft = 1/6 of 24 ft.

Hinglish: *Pehla bounce 24 ka 3/4 = 18 ft. Har baar 3/4 - GP with r = 3/4. 24 ka 1/6 = 4 ft. 6th bounce 4.27 (abhi bhi upar), 7th bounce 3.20 - ab 4 ke neeche. Ball ki story bhi GP hai, bas shrinking wali.*

IMP Q8 | 31st term from t11 and t16 (T9)

Q8. Find the 31st term of an AP whose 11th term is 38 and whose 16th term is 73.

Step 1. a + 10d = 38 and a + 15d = 73.

Step 2. Subtract: 5d = 35, so d = 7.

Step 3. Back-substitute: a = 38 - 10 × 7 = 38 - 70 = -32.

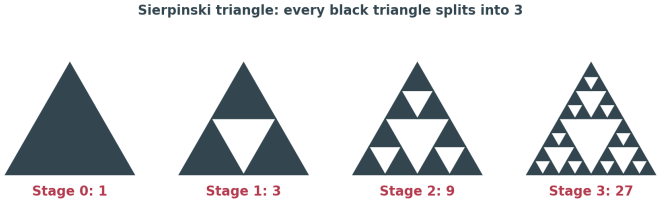
Step 4. t_{31} = a + 30d = -32 + 30 × 7 = -32 + 210 = 178.

Answer: 178.

Hinglish: *11th aur 16th ke beech 5 steps = 5d = 35, d = 7. a negative nikla (-32) - koi problem nahi, AP negative se shuru hokar chadhti hai. 31st = 178.*

IMP Q9 | Sierpinski triangle counts and areas (T13)

Q9. The figure shows Stages 0 to 3 of the Sierpiński triangle, built from an equilateral triangle by repeatedly joining midpoints and removing the central triangle. (i) How many black triangles are there at Stages 0 to 3? (ii) Predict Stages 4 and 5 and find the rule for Stage n. (iii) If the Stage-0 black area is 1 square unit, what is the black area at Stage n, and what happens to it as n grows?



Step 1. (i) Stage 0: 1 black triangle; Stage 1: 3; Stage 2: 9; Stage 3: 27 - every black triangle is replaced by 3 smaller ones.

Step 2. (ii) Stage 4: 81; Stage 5: 243. Rule: at Stage n the count is 3^n - a GP with a = 1, r = 3 indexed from Stage 0.

Step 3. (iii) Each stage keeps 3 of every 4 equal pieces, so the area multiplies by 3/4: area at Stage n = $(3/4)^n$ (1, 3/4, 9/16, 27/64, ...).

Step 4. As n grows, $(3/4)^n$ heads to 0 - the number of triangles explodes while the total black area melts away.

Answer: (i) 1, 3, 9, 27 (ii) 81 and 243; count at Stage n = 3^n (iii) area = $(3/4)^n$, which tends to 0.

Hinglish: *Count har stage par 3 se multiply: 1, 3, 9, 27, 81, 243 - Stage n pe 3^n. Area har baar 3/4 ho jaata hai: $(3/4)^n$. Ek cheez badhti hai, doosri ghatti hai - fractal ka jaadu. Stage 0 se shuru - exponent mein +1 ka farak yaad rakho.*

IMP Q10 | Three numbers in AP: sum and product (T9)

Q10. Three numbers are in AP. Their sum is 27 and their product is 504. Find the numbers.

Step 1. Let the numbers be 9 - x, 9, 9 + x (middle term = average = 27/3 = 9).

Step 2. Product: $(9 - x) \times 9 \times (9 + x) = 9(81 - x^2) = 504$.

Step 3. $81 - x^2 = 56$, so $x^2 = 25$, x = 5 (or -5, which only reverses the order).

Step 4. Numbers: 4, 9, 14. Check: sum 27 ✓; product $4 \times 9 \times 14 = 504$ ✓.

Answer: 4, 9, 14 (or the same three in reverse order).

Hinglish: *Teen AP numbers ka middle = average = 9. Baaki do ko 9 - x, 9 + x likho - sum apne aap sambhal jaata hai. Product equation se $x^2 = 25$. x = ±5 dono same set dete hain, bas order ulta.*

IMP Q11 | GP: sum 13/12, product -1 (T12)

Q11. The sum of the first three terms of a GP is 13/12 and their product is -1. Find the common ratio and the three terms.

Step 1. Let the terms be a/r, a, ar. Product = (a/r) × a × (ar) = a^3 = -1, so a = -1.

Step 2. Sum: a(1/r + 1 + r) = 13/12, so -(1/r + 1 + r) = 13/12.

Step 3. Thus 1/r + r = -13/12 - 1 = -25/12.

Step 4. Multiply by 12r: 12 + 12r^2 = -25r, i.e. 12r^2 + 25r + 12 = 0.

Step 5. Split the middle term: 12r^2 + 16r + 9r + 12 = 4r(3r + 4) + 3(3r + 4) = (3r + 4)(4r + 3) = 0.

Step 6. r = -4/3 or r = -3/4.

Step 7. For r = -3/4: terms 4/3, -1, 3/4 (sum 16/12 - 12/12 + 9/12 = 13/12 ✓, product -1 ✓).

Step 8. For r = -4/3: terms 3/4, -1, 4/3 - the same three numbers in reverse order.

Answer: r = -3/4 (or -4/3); terms are 4/3, -1, 3/4 in some order.

Hinglish: *Symmetric writing se product = a^3 = -1, matlab middle term -1. Quadratic 12r^2 + 25r + 12 middle-term split se (3r + 4)(4r + 3). Do r aaye par dono same teen numbers dete hain, bas order ulta. Boss-level question - ye kar liya toh chapter ka raja tum ho.*

8.5 Rapid-Fire MCQs (12)

Rapid fire: option padho, dimaag mein answer banao, phir Answer line se check karo. Galat hua toh Answer line ka logic padho - wahi 1 line exam mein marks bachayegi.

1. Which of the following is an arithmetic progression?

(A) 1, 2, 4, 8, 16 **(B)** 2, 5, 8, 11, 14 **(C)** 1, 4, 9, 16, 25 **(D)** 1, 1/2, 1/4, 1/8

Ans: (B) · 2, 5, 8, 11, 14 has a constant difference of 3 between consecutive terms - the definition of an AP. The others are a GP (doubling), the squares (differences 3, 5, 7 grow) and a halving GP.

2. The common difference of the AP 5, 9, 13, 17, ... is:

(A) 4 **(B)** 5 **(C)** 9 **(D)** -4

Ans: (A) · d = t_2 - t_1 = 9 - 5 = 4 (and 13 - 9 = 4 confirms it).

3. The 10th term of the AP 7, 11, 15, 19, ... is:

(A) 43 **(B)** 47 **(C)** 39 **(D)** 51

Ans: (A) · a = 7, d = 4. t_{10} = 7 + 9 × 4 = 7 + 36 = 43.

4. Which term of the AP 4, 9, 14, 19, ... is 109?

(A) 21st **(B)** 22nd **(C)** 20th **(D)** 23rd

Ans: (B) · a = 4, d = 5. Solve 4 + (n - 1) × 5 = 109: (n - 1) × 5 = 105, n - 1 = 21, n = 22.

5. The sum 1 + 2 + 3 + ... + 15 equals:

(A) 120 **(B)** 105 **(C)** 240 **(D)** 135

Ans: (A) · $S_{15} = 15 \times 16/2 = 15 \times 8 = 120$.

6. The 8th triangular number is:

(A) 28 **(B)** 36 **(C)** 45 **(D)** 24

Ans: (B) · $T_8 = 8 \times 9/2 = 36$ (also 1 + 2 + ... + 8).

7. The common ratio of the GP 2, -6, 18, -54, ... is:

(A) 3 **(B)** -3 **(C)** -2 **(D)** 1/3

Ans: (B) · $r = t_2/t_1 = -6/2 = -3$. Check: 18/(-6) = -3 ✓.

8. The 5th term of the GP 3, 12, 48, ... is:

(A) 192 **(B)** 768 **(C)** 144 **(D)** 576

Ans: (B) · a = 3, r = 4. The 5th term is $ar^4 = 3 \times 4^4 = 3 \times 256 = 768$. (The exponent is 4, not 5 - the first term uses r^0.)

9. The next term of the sequence 1, 2, 3, 5, 8, 13, ... is:

(A) 18 **(B)** 21 **(C)** 26 **(D)** 16

Ans: (B) · Each term is the sum of the previous two (Virahānka-Fibonacci): 8 + 13 = 21.

10. A sequence has $t_1 = 2$ and $t_n = t_{n-1} + 5$. Its 6th term is:

(A) 27 **(B)** 32 **(C)** 30 **(D)** 22

Ans: (A) · This is an AP with a = 2, d = 5: $t_6 = 2 + 5 \times 5 = 27$.

11. The sequence 1, 1/2, 1/4, 1/8, ... continues with next term:

(A) 1/16 **(B)** 1/10 **(C)** 1/12 **(D)** 1/6

Ans: (A) · Common ratio r = 1/2, so the next term is 1/8 × 1/2 = 1/16.

12. In the Sierpinski triangle (Stage 0 = 1 black triangle), the number of black triangles at Stage 4 is:

(A) 27 **(B)** 81 **(C)** 243 **(D)** 12

Ans: (B) · The count is 3^n at Stage n: Stage 4 gives $3^4 = 81$.

Exam Morning Power Page (Subah Sirf Ye Page)

Subah exam se pehle poora book nahi - sirf ye page. Ye whole syllabus ke 12 sabse killer formulas hain jo har section mein kaam aate hain, plus exam-hall ke 8 rules. Is page ko 3 baar padho, gehri saans lo, aur exam dené jao. Tum ready ho.

Killer formula	Ch	Kahan kaam aayega
Distance $\sqrt{((x_2-x_1)^2 + (y_2-y_1)^2)}$ + triples (3,4,5)	Ch 1	Coordinates se har faasla; triple dikhe toh 2-second answer.
y = ax + b (a = rate, b = fixed)	Ch 2	Bill/pattern/age - sab isme; slope compare = steeper/parallel.
Recurring decimal: block/99...9; mixed: (all−non-repeat)/(9..0..)	Ch 3	0.454545 = 45/99; 0.4777... = 43/90. Denominator $2^m \times 5^n$ = terminating.
$a^m \times a^n = a^{m+n}$; $a^{m/n} = \sqrt[n]{(a^m)}$; same base → equate	Ch 3	$8 = 2^3$, $1/81 = 3^{-4}$ banao, powers barabar rakho.
$(a \pm b)^2$; $a^2-b^2 = (a+b)(a-b)$; (x+a)(x+b)	Ch 4	98^2, 51×49, factorisation - sab inhi se.
Crown: a+b+c = 0 → $a^3+b^3+c^3 = 3abc$	Ch 4	Zero-sum dikha, 3abc likho, marks pakke.
$x + 1/x = s \rightarrow s^2-2 \rightarrow s^3-3s$ → chain	Ch 4	$x-1/x = d$ ho toh d^2+2, d^3+3d.
Heron: s → 3 differences → multiply → √; right triangle → ½·leg·leg	Ch 6	Pehle 5-12-13 check karo - Heron skip ho sakta hai.
Arc $\ell = (\theta/360)2\pi r$; Sector = $(\theta/360)\pi r^2 = \frac{1}{2}\ell r$; Ring = $\pi(R+r)(R-r)$	Ch 6	π choice: 7 ka multiple → 22/7, warna 3.14.
P(E) = fav/total; P(not E) = 1−P(E); at least one → 1 − P(none)	Ch 7	2 dice = 36, 3 coins = 8, cards: 4 ace / 12 face / 26 red / 13 suit / 9 numbered.
AP: $t_n = a + (n-1)d$; Sum = n(n+1)/2; GP: $t_n = ar^{n-1}$	Ch 8	Which term → n natural? Sierpinski: 3^n, $(3/4)^n$.
Virahanka: $V_n = V_{n-1} + V_{n-2}$ (1,2,3,5,8,13,...)	Ch 8	Pichhle do ka sum - 1 mark ka pakka question.

Exam Hall ke 8 Rules

- **Figure banao** - coordinate geometry, area, probability (grid) sab mein 2-inch figure half confusion khatam karta hai.
- **Given values kabhi random nahi hote** - a+b, ab diye hain toh bridge lagana hi hai; s diya hai toh Heron.
- **π ka sahi choice:** 7 ka multiple → 22/7; warna 3.14. Galat choice = pura question galat.
- **Units likho** - cm, cm^2, m/s. Unit chhoda toh half mark kata jata hai.
- **MCQ strategy:** options se value daal ke check karna allowed hai - 98^2 type mein options khud bata denge.
- **Steps mat chhodo:** identity ka naam / formula likhna = method marks, chahe calculation slip ho jaye.
- **Last 10 minutes:** sirf CHECKING - units, signs (minus wale terms), aur "at least one" ke complements dobara dekho.

- **Trap alarm:** $(a-b)^2$ ka last term PLUS hota hai; HT aur TH alag outcomes; without replacement mein denominator bhi shrink hota hai.

Subah ka 10-Minute Drill (Exam se theek pehle)

1. Exam Morning Power Page ke 12 formulas **bol ke ek baar likho** - haath se likhne se muscle memory banti hai.
2. Apne **starred MCQs aur galtiyan** dohrao - sirf wahi, teen minute. Naya kuch nahi, sirf apni mistakes.
3. Trap alarms zor se bolo: $(a-b)^2$ ka last term PLUS, HT ≠ TH, without replacement mein denominator shrink, π ka sahi choice.
4. Triples aur squares ghoshit karo: 3-4-5, 5-12-13, 8-15-17, 7-24-25; squares $11^2 = 121$ se $20^2 = 400$ tak.
5. Geometry formulas bol ke revise karo - Heron ka s, sector ka $\theta/360$, ring ka $(R+r)(R-r)$. Awaaz nikalne se yaad pakka hota hai.
6. Gehri saans lo aur bolo: **"Maine 172 solved examples kiye hain - paper bas unka naya version hai."** Ab jaao, 80/80 tumhara hai.

> **Aakhri baat**
> Tumhare paas poora syllabus, saare tricks aur 172 solved examples hain. Kal bas wahi dikhao jo tumne practice kiya hai. Sheet tumhare saath hai, formula tumhare dimaag mein - ab sirf likhne ki deri hai. All the best - **80/80 is yours.**